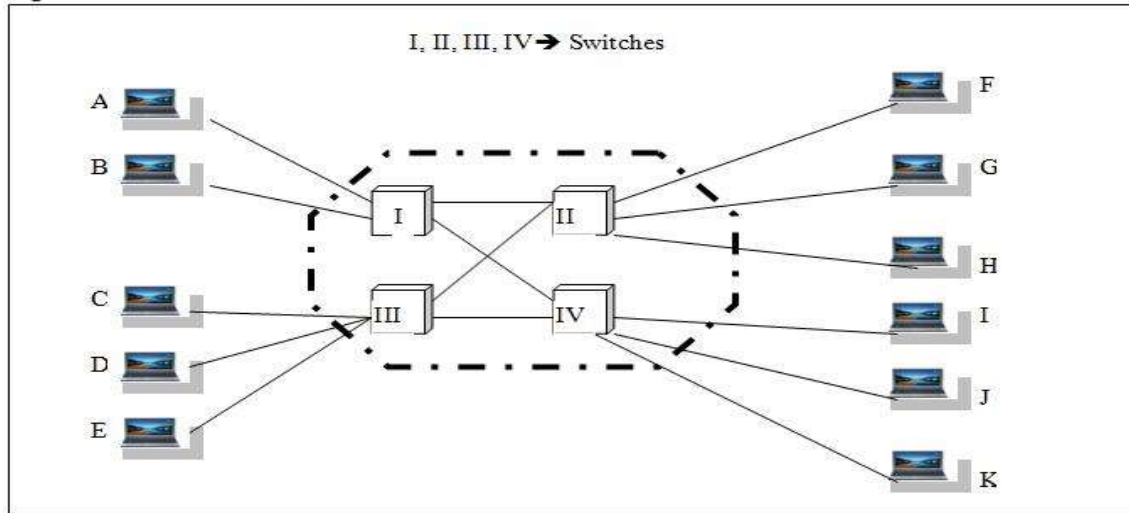


Switching:

Whenever we have multiple devices to connect with each other, the better solution is switching. A switched network consists of a series of interlinked nodes called **switches**.

Switches are hardware/software devices that are capable of creating the temporary connections between two or more devices link to the switch.

Figure: *Switched Network*

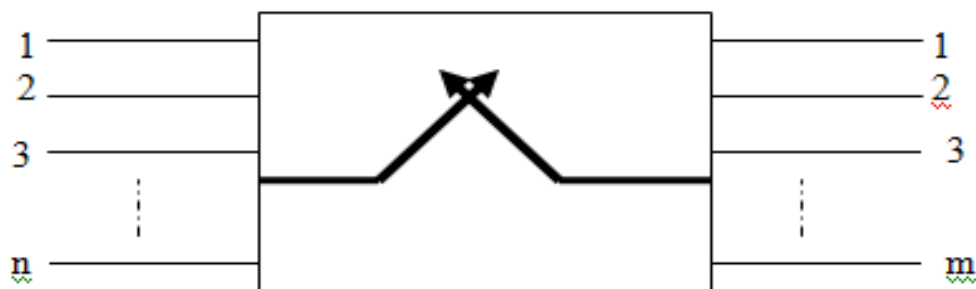


There are three methods for switching.

- I) Circuit switching
- II) Packet Switching
- iii) Message Switching

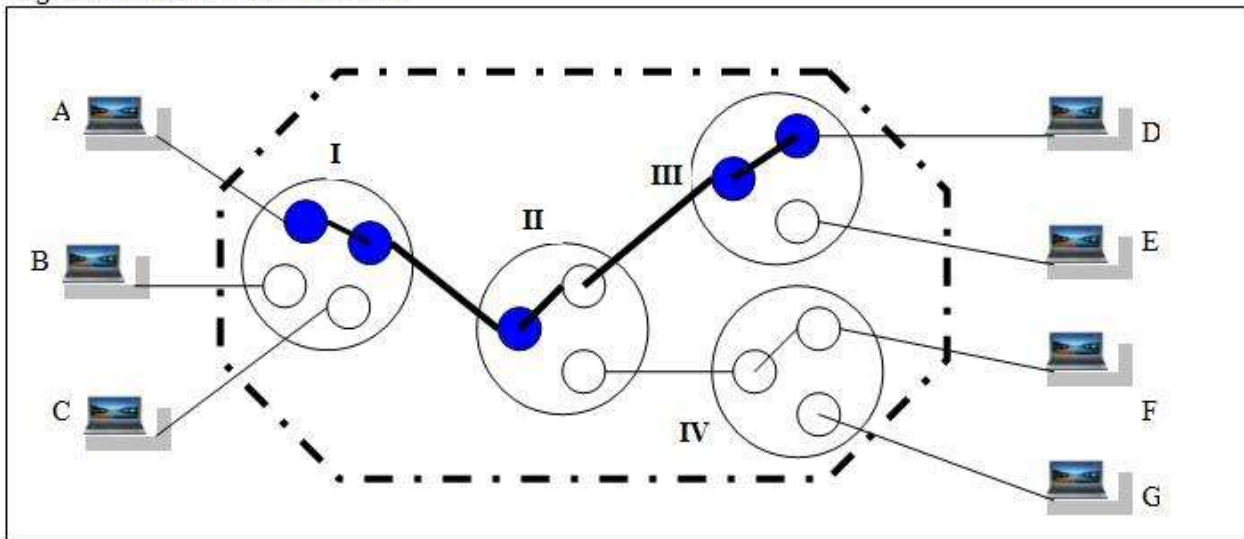
I) Circuit Switching:

Circuit switching creates a direct physical connection between two devices such as phones. A circuit switch is a device with n inputs and m outputs that creates a temporary connection between an input link and an output link. The number of inputs may have different with the number of outputs.



A Circuit Switch

Figure: *Circuit-Switched Network*



Circuit Switching use either of two technologies:

- 1) Space-Division Switches
- 2) Time-Division Switches

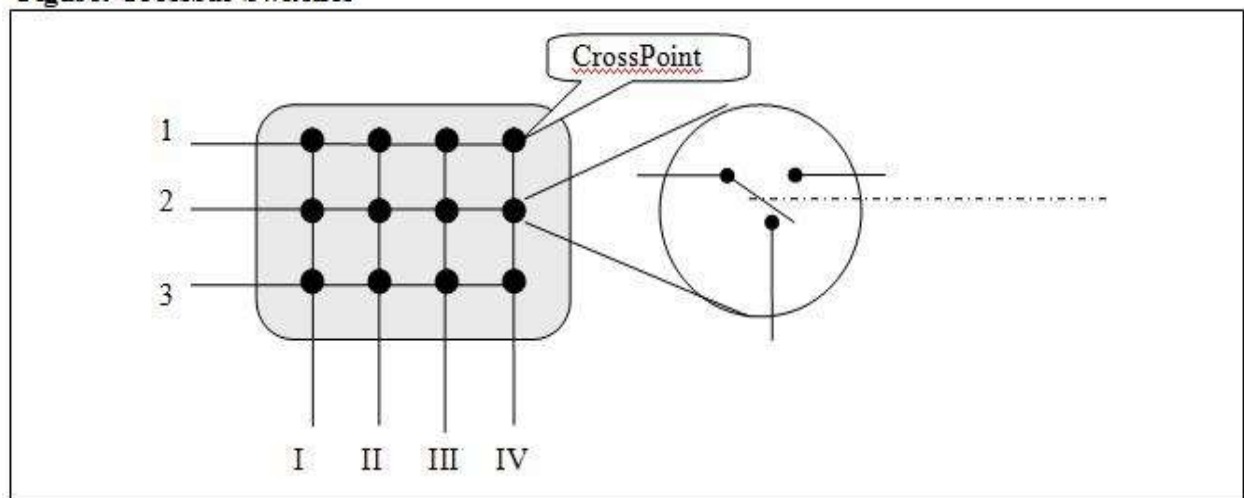
1) Space-Division Switches:

In space-division switching, the path for each communication in the circuit are separated from each other. There are many technologies used to design the **Space Division Switches**.

a) Crossbar Switches:

A **crossbar switch** connects n inputs to m outputs in a grid using transistor at each cross point. So for connecting n inputs and m outputs there are $n \times m$ cross points required.

Figure: Crossbar Switches

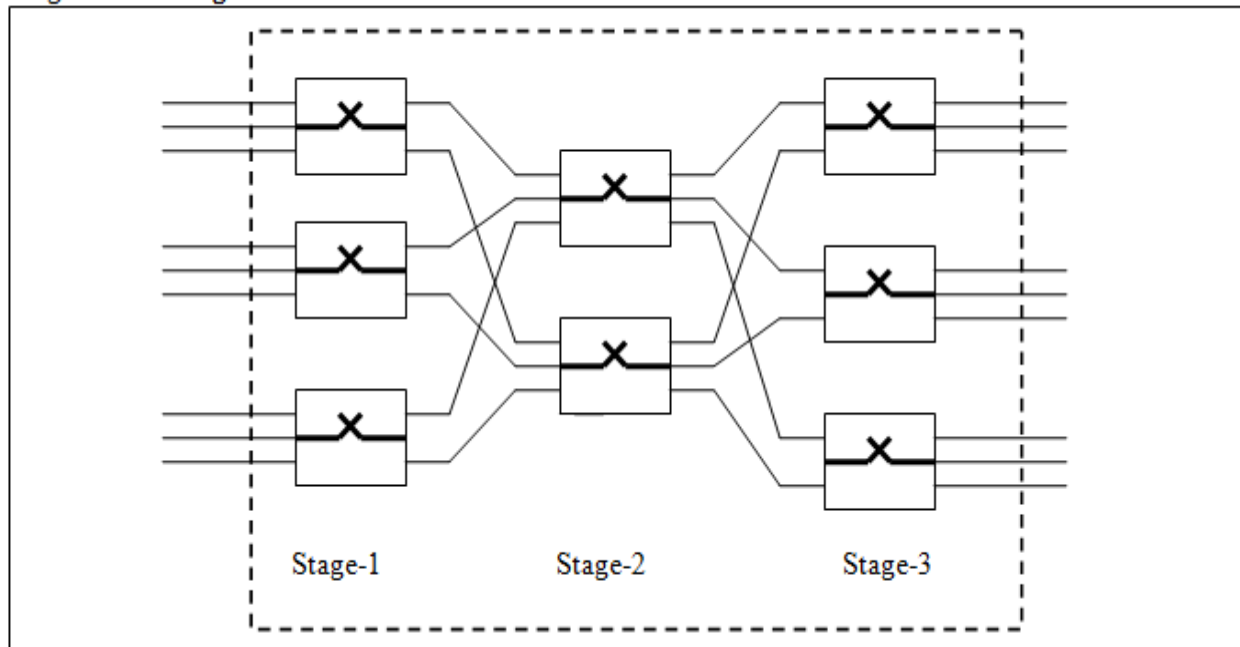


This factor makes the cross impractical because readings show that in practice fewer than 25% of the cross points are in use at a given time. The rest are ideal.

b) Multistage Switches:

The solution to reduce the number of the crossbar switch is to use multistage switches, which combine crossbar switches in several stages.

Figure: *Multistage Switch*



The design of multistage switch depends on the number of stages and the number of switches requires in each stage. Normally middle stages have fewer switches than the first and last stage. Let us compare the number of cross points in a 9-by-9 single stage crossbar switch with the 15-by-15 multistage switch.

In a single stage switch, we need 81 cross points (9×9).

In the multistage switch, we need

- ❖ Three, first stage switches, each with 6 cross points (3×2), total of 18 cross-points at the first stage.
- ❖ Two, second stage switches, each with 9 cross points (3×3), total of 18 cross-points at the first stage.
- ❖ Three third stage switches, each with 6 cross points (3×2), total of 18 cross-points at the first stage.

So in multistage switching we need only 54 total crosspoints which is 34% less than single stage switching.

2) Time-Division Switches:

Time-division switching uses time-division multiplexing to achieve switching. There are two popular methods used in time-division multiplexing:

- a) Time-slot Interchange
- b) TDM Bus

a) Time-Slot Interchange (TSI):

In **TSI**, we insert a device called **Time-slot interchange (TSI)**. A TSI changes the ordering of the slots based on the desired connections. In the figure, it changes the order of data from A, B, C, D to C, D, A, B. Now at the receiver end demultiplexer separates the slots, it passes them to the proper output.

Figure: *Time-Division Multiplexing without TSI (No Switching)*

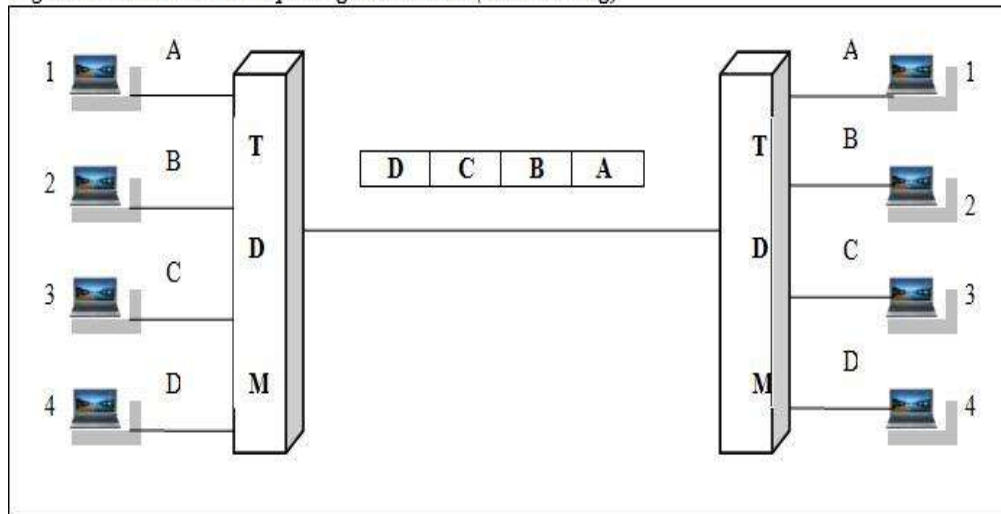
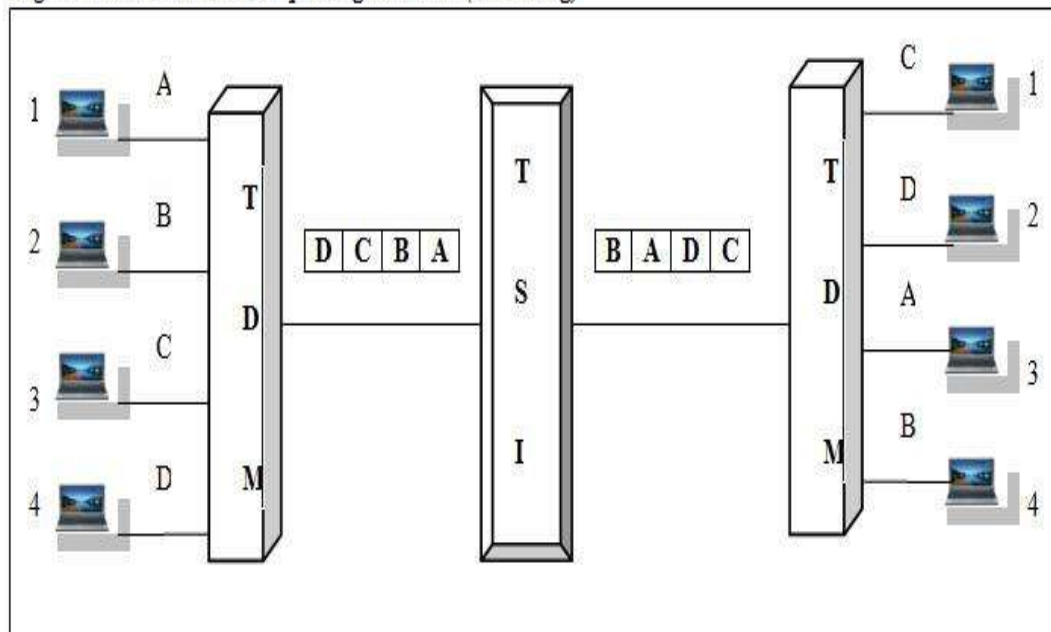


Figure: *Time-Division Multiplexing with TSI (Switching)*



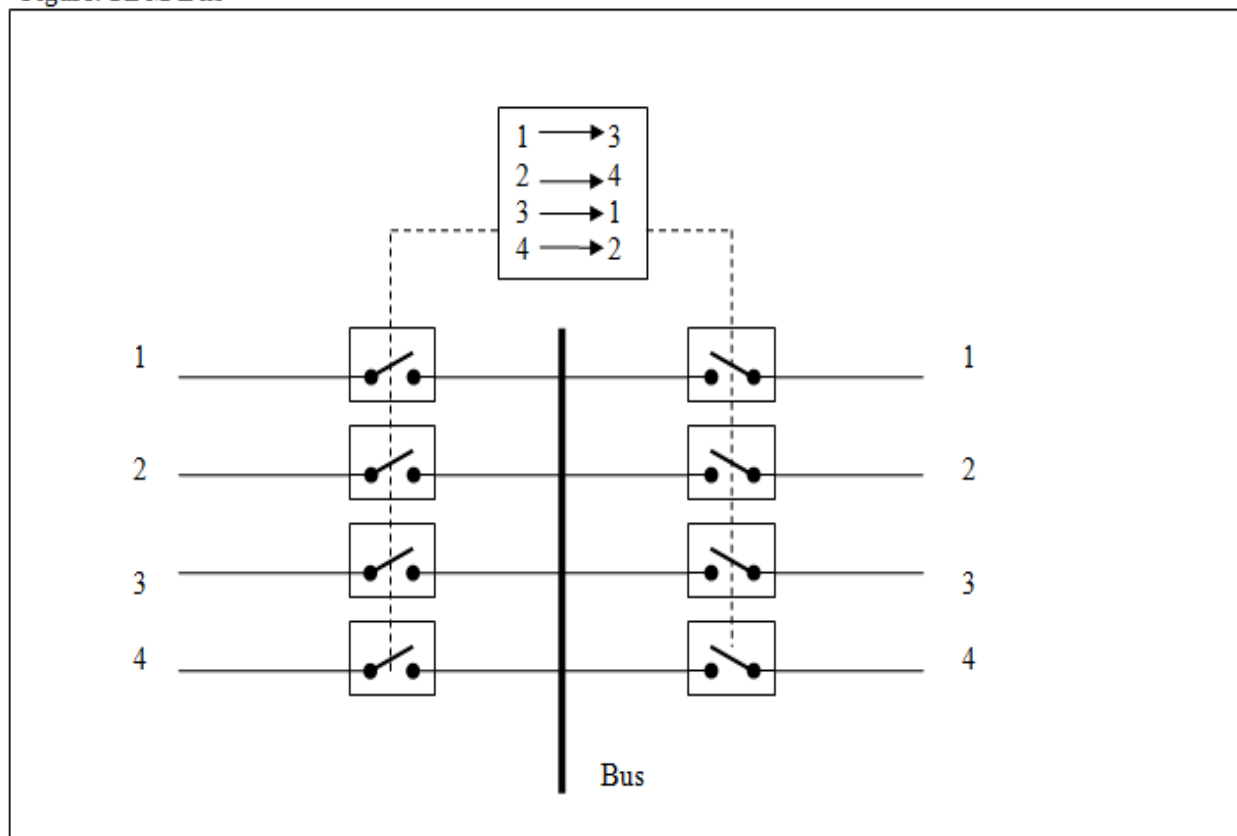
TSI consists of RAM (random access memory) with several memory locations. The size of each location is the same as the size of a single time slot. The RAM fills up with incoming data from

time slots in the order received. Slots are then sent out in an order based on the decisions of control unit.

b) TDM Bus:

In **TDM Bus**, the input and output lines are connected to a high speed bus through input and output gates. During the data transfer one input gate is closed for a time slots, during the same time slot, only one output gate is also closed. This pair of gates allows transferring of data from one specific input line to one specific output line using bus. The control unit opens and closes the gate accordingly to switching need. For example: in the figure, at the first time slots the input gate 1 and output gate 3 will be closed; during the second time slot, input gate 2 and output gate 4 will be closed; and so on.

Figure: *TDM Bus*



II) Packet Switching:

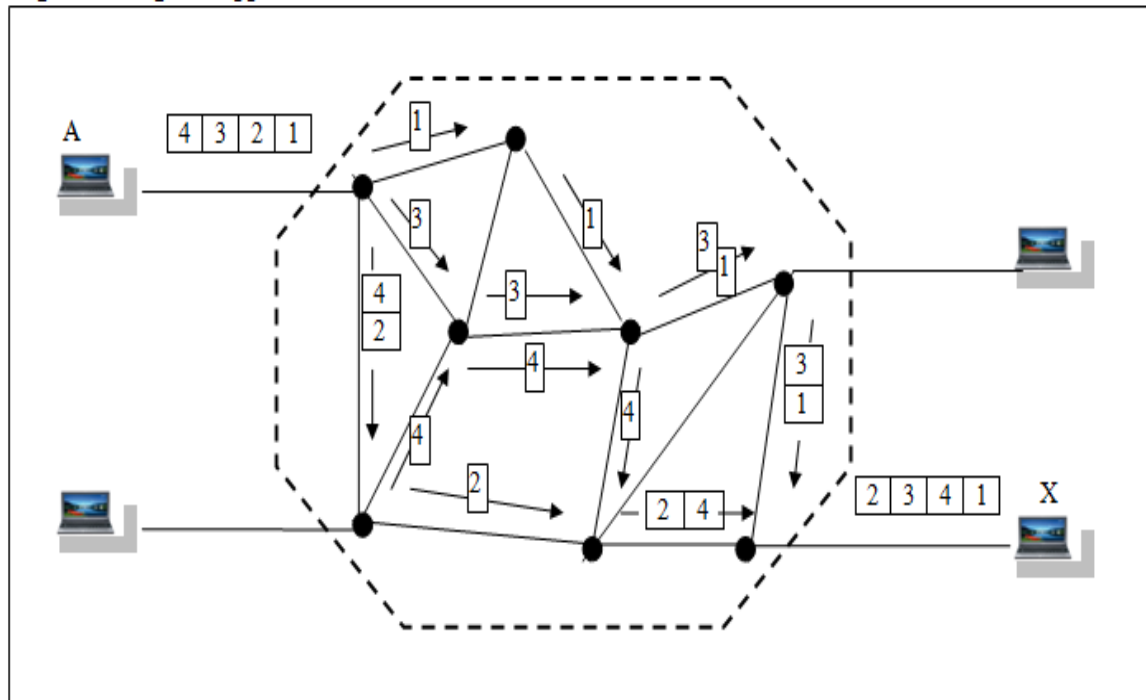
In a **packet switched network**, data are transmitted in discrete units of variable length blocks called **packets**. Longer transmissions are broken up in to multiple packets. Each packet contains not only data but also a header with control information (such as priority codes, source and destination address). The packets are sent over the network node to node. At each node, the packet is stored then routed according to the information in its header. There are two popular approaches for packet switching

- 1) Datagram Approach
- 2) Virtual Circuit Approach

1) Datagram Approach:

In **datagram approach**, each packet is treated independently from all others. Packets in this technology are referred to as **datagram**. In this approach the datagram reach to their destination out of order. It is the responsibility of the transport layer to reorder the datagram before passing them to destination port.

Figure: Datagram Approach



Virtual

2) Virtual Circuit Approach:

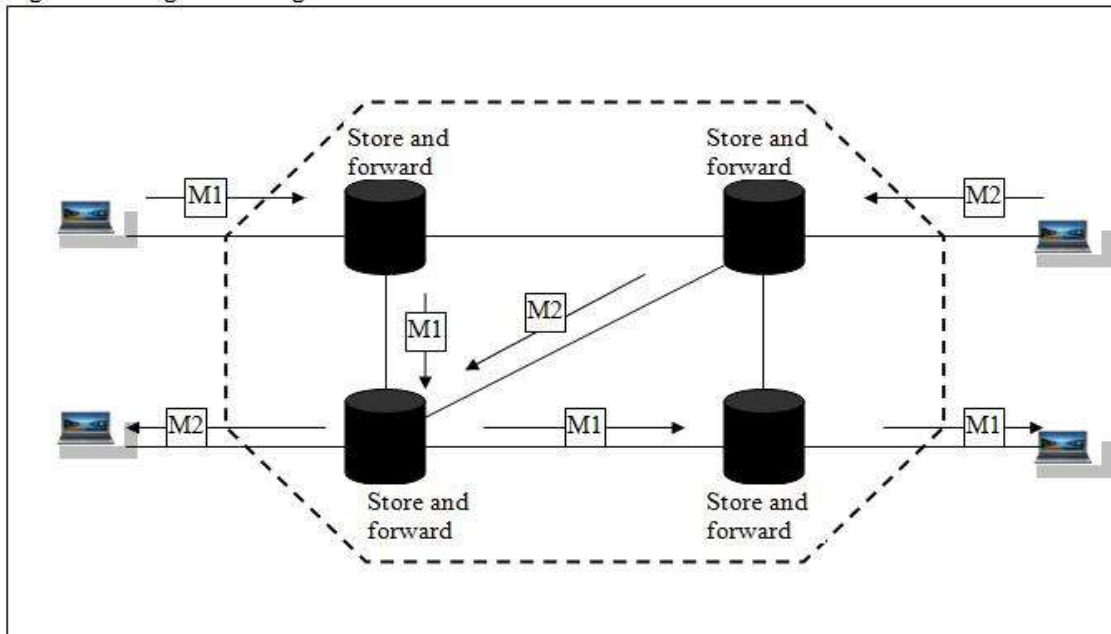
In **virtual circuit approach**, the relationship between all packets belonging to a message or session is preserved. A single route is chosen between sender and receiver at the beginning of the session. When data are sent, all packets of the transmission travel one after another along that route only.

III) Message Switching:

In **message switching**, a node receive a message, store it until the appropriate route is free and then send forward.

Note that in message switching, the message are stored and relayed from secondary storage (disk), while in packet switching the packet are stored and forwarded from primary storage (RAM).

Figure: Message Switching



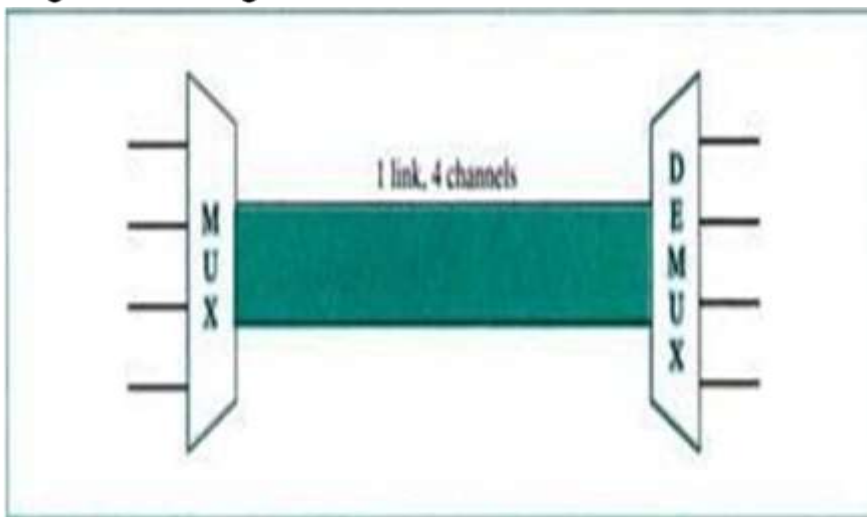
Multiplexing:

Whenever the bandwidth of the medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.

Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single link.

In multiplexed system, 1 line share the bandwidth of n link

Figure: Dividing a link in to 4 channel



The four line on the left direct their transmission streams to a multiplexer (**MUX**), which combines them in to single stream (many to one). At the receiving end that stream is fed in to a demultiplexer (**DEMUX**), which separates the stream back in to its component transmission (one to many), and direct them in to their corresponding lines.

In figure, the word link refers to the physical path and channel refers to the portion of a link that carries a transmission between a given pair of line.

There are three techniques for multiplexed the signals

1) Frequency division multiplexing

2) Wave division multiplexing

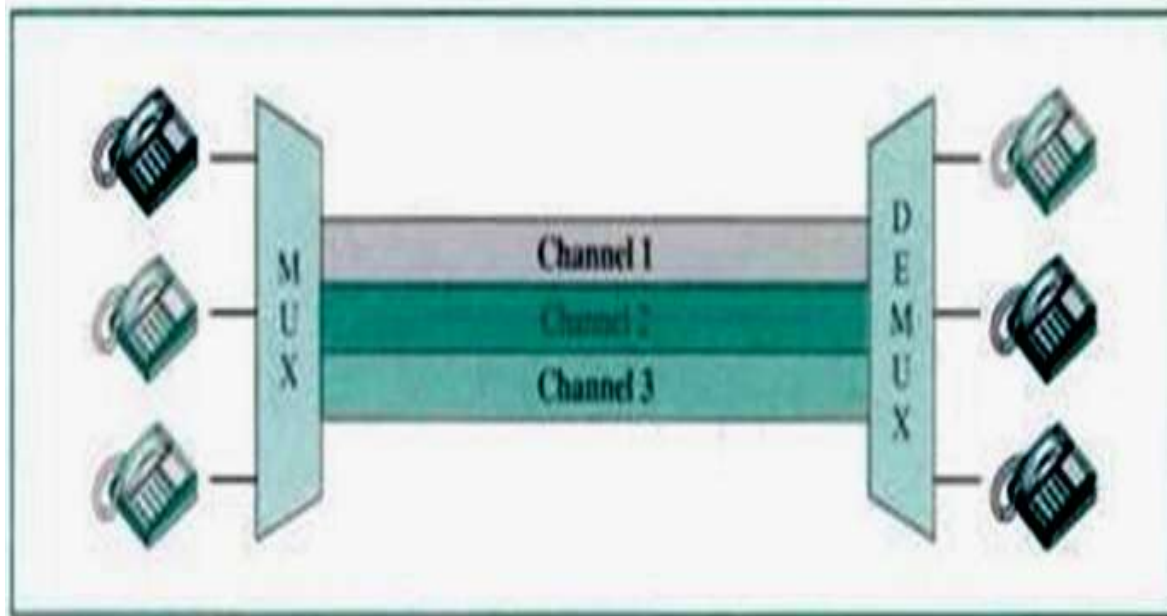
3) Time division multiplexing

The first two techniques are used for analog signals and third one use for digital signals.

1) Frequency division multiplexing:

In **FDM**, signals generated by each sending device, modulate at different carrier frequencies. These modulate signals are then combined in to a single composite signal that can be transported by the link. Carrier frequency are sufficient to accommodate the modulated signal. Channel must be separated by unused bandwidth to prevent signal form overlapping.

Figure: FDM



Multiplexing Process:

FDM is an analog process, here we use telephone as an input device. Each telephone generates a signal of similar frequency range. In side the multiplexer, these similar signals are modulated on to different carrier frequencies (f_1 , f_2 , f_3). The resulting modulated signals are then combined in to a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.

Figure: multiplexing

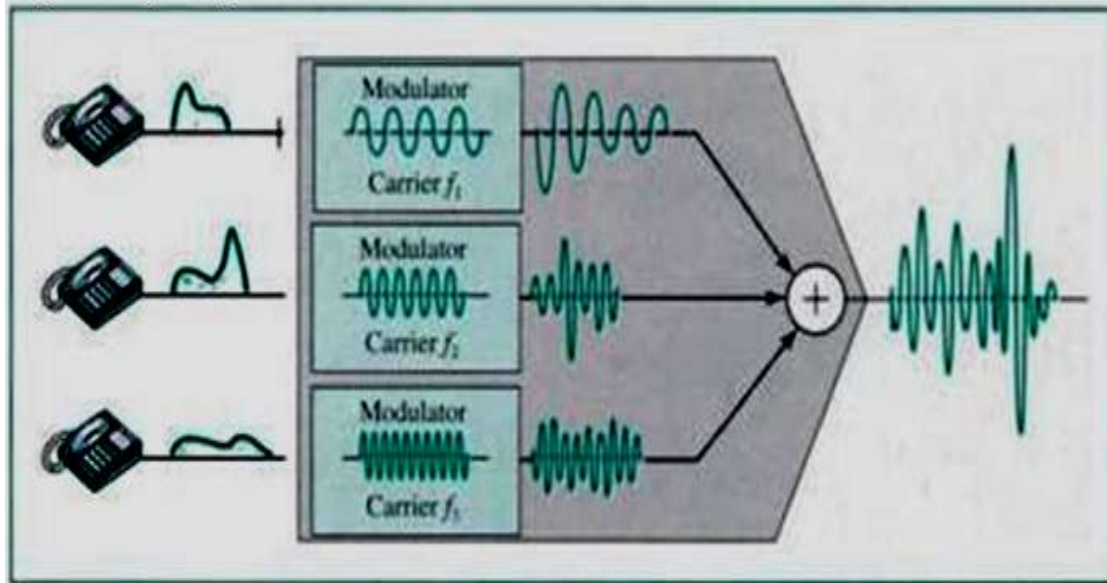
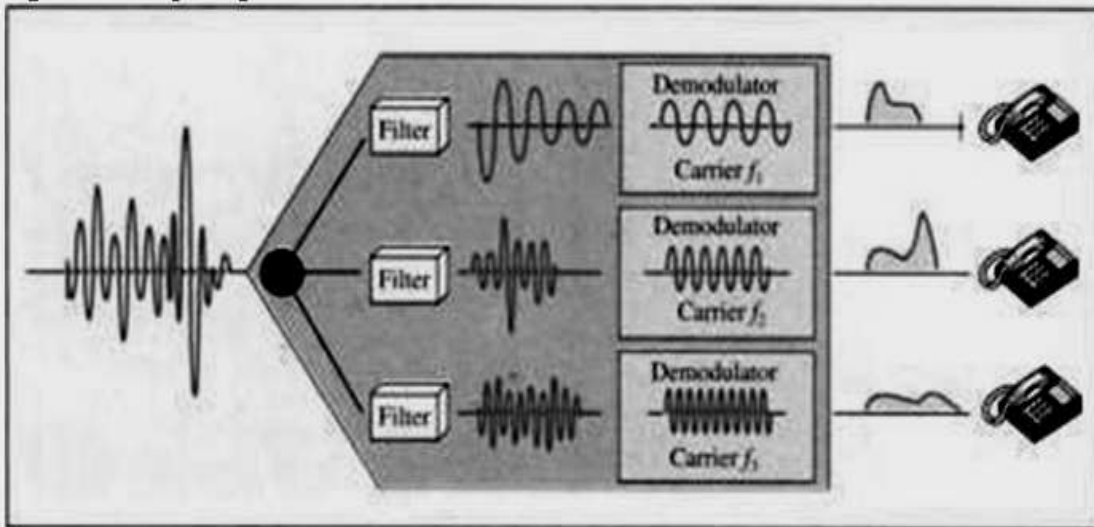


Figure: Demultiplexing



Demultiplexing Process:

The demultiplexer uses a series of filters to decompose the multiplexed signals into its component signals. The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the waiting receiver.

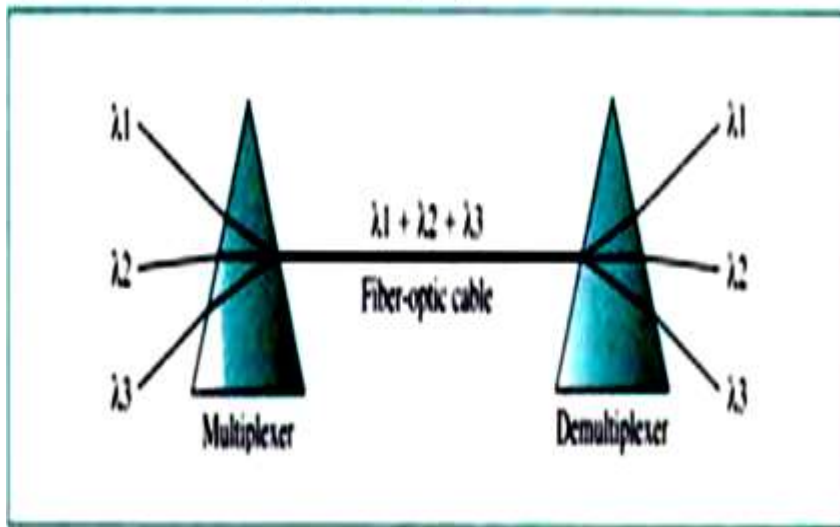
2) Wave division multiplexing:

WDM is designed to use the high data rate capability of fiber optic cable. The optical fiber data rate is higher than the data rate of metallic transmission cable.

WDM uses the same concept as the FDM except that the multiplexing and demultiplexing involve optical signals transmitted through fiber optic channels.

Combining and splitting of light sources are easily handled by a prism. Using this technique, a multiplexer can be made to combine several input beams of light, each containing a narrow band of frequencies into one output beam of a wider band of frequency. A demultiplexer can also be made to reverse the process.

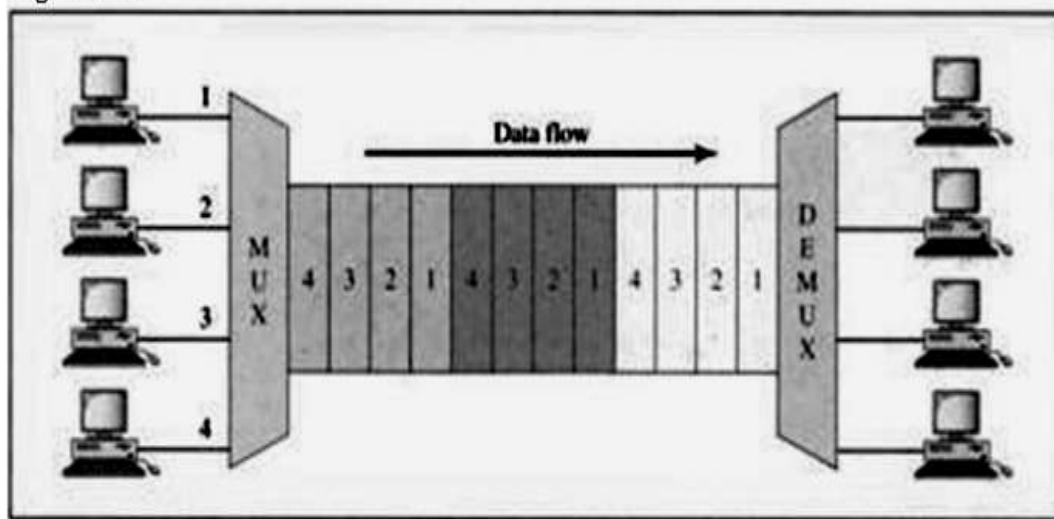
Figure: Prism in WDM multiplexing and demultiplexing



3) Time Division Multiplexing:

Time Division multiplexing (TDM) is a digital process. Instead of sharing a portion of bandwidth as in FDM, time is shared. Each connection occupies a portion of time in the link

Figure: TDM



TDM can be implemented in two ways.

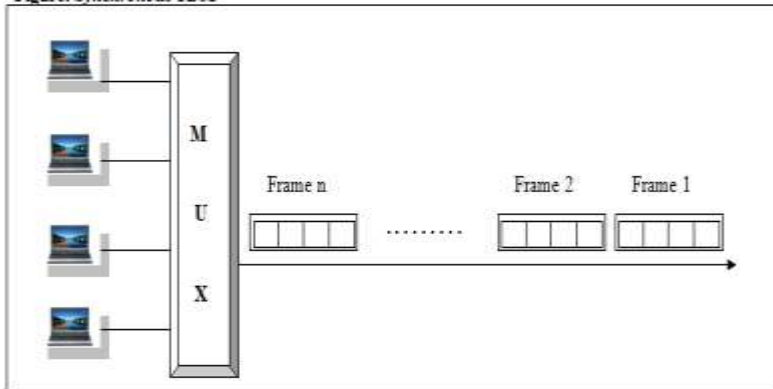
a) **Synchronous TDM**

b) **Asynchronous TDM**

a) **Synchronous TDM:**

In synchronous TDM the multiplexer allocates exactly the same time slots to each device at all times, whether or not a device has anything to transmit. So if there are n device then the number of time slots are also n . For Example, time slot A is assigned to device A alone and no one can use it. Each time its allocated times comes up, a device has the opportunity to send a portion of its data. If a device is unable to transmit or does not have data to send, its time slot remains empty.

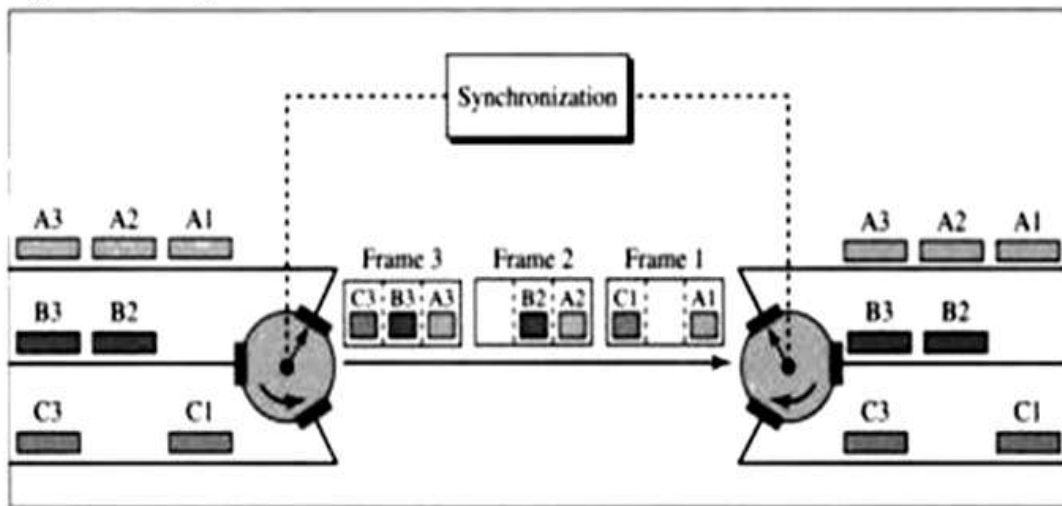
Figure: Synchronous TDM



Interleaving:

Synchronous TDM can be understood as two fast rotating switches, one on the multiplexing side and the other on the demultiplexing side. The switch is synchronized and rotates at the same speed, but in opposite directions. On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit on the path. This process is called interleaving. On the demultiplexing side, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from that path.

Figure: Interleaving



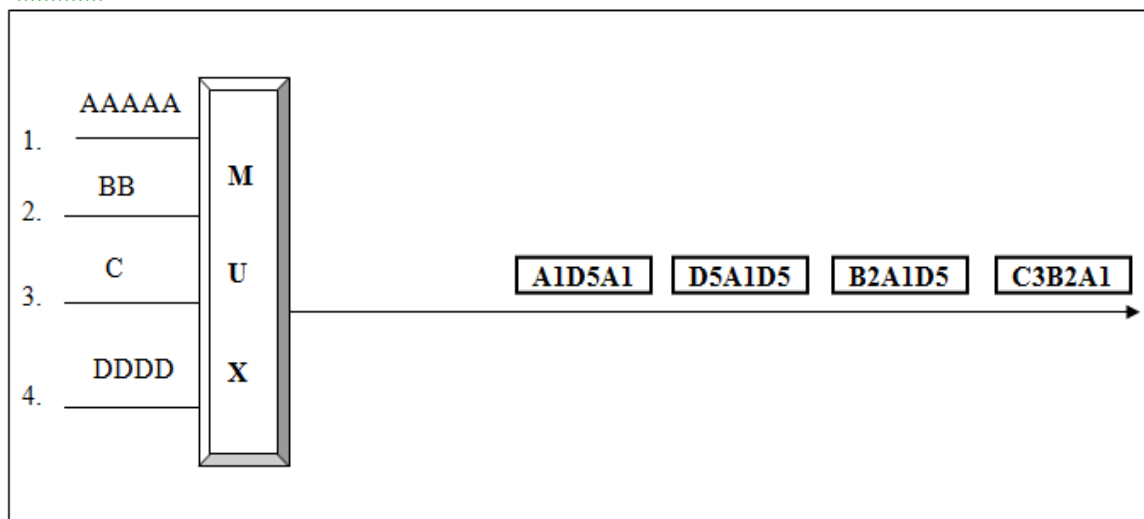
Synchronization:

For synchronization bits are usually added to beginning of each frame. These bits called **framing bits** follow a pattern frame to frame. So that the demultiplexer can be synchronized with incoming bit pattern. in most cases, this synchronization information consists of 1 bit per frame, alternating 0 and 1.

b) Asynchronous TDM:

Synchronous TDM does not guarantee that the full capacity of the link is used. Because whenever the connecting device has nothing to transmit, the corresponding time slot is empty and that much of the path is wasted. **Asynchronous TDM** is designed to avoid this type of wastage. In asynchronous TDM if we have n input lines then the frame contain no more than m slots where $m < n$.

Figure : Asynchronous TDM



Addressing and overhead:

In asynchronous TDM bit in the time slot may be at first from device 1, and in next time slot may be at third. So for the synchronization in Asynchronous TDM, each time slot carry an address telling the demultiplexer how to direct the data. This address is only for local use only. it is attached by multiplexer and discards by the demultiplexer once it has been read. So adding the address bit to each time slot increase the overhead of an asynchronous system and some what limits the potential efficiency.

Signals:

When data moves from one node to another, data must be transformed in to electromagnetic signals.

Both data and the signals that represent them can take either analog or digital form. So signals can be of two types.

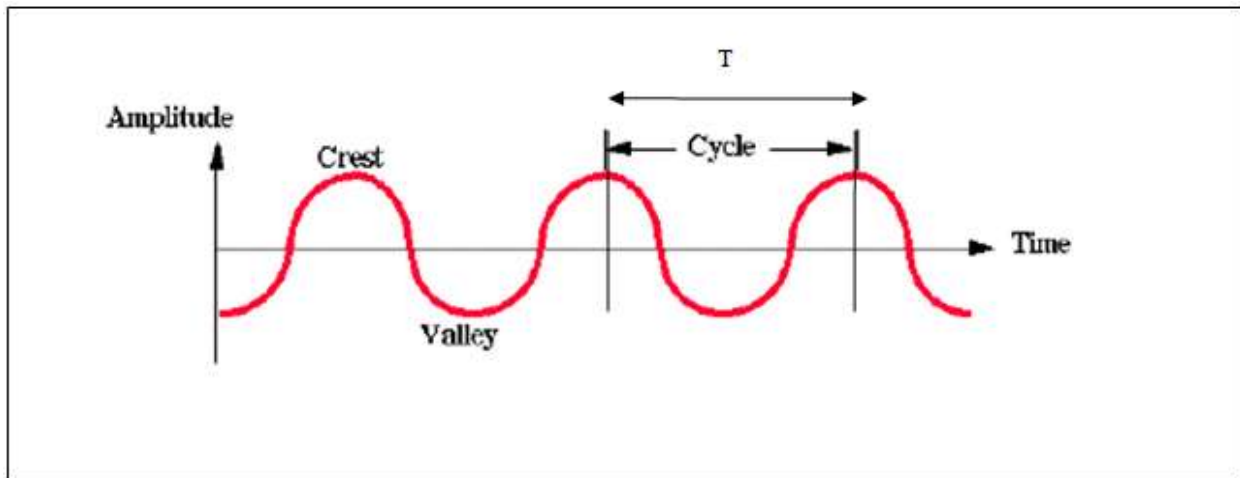
1. Analog signal
2. Digital signal

Analog :

Analog refers to something that is continuous- a set of specific points of data and all possible points between.

Analog signal:

An analog signal is a continuous wave form that changes smoothly over time. An analog signal can have any value between starting and end range. So analog signal includes infinite number of value along its path.



Analog signals can be classified as simple or composite. A simple analog signal or a **sine wave**, can not be decomposed in to simpler signals. A composite analog signal is composed of multiple sine waves.

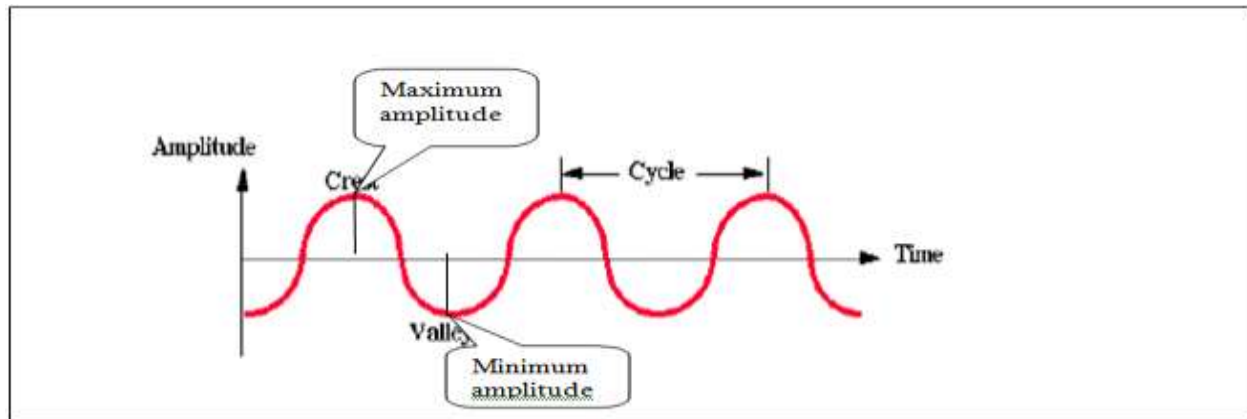
Simple analog signal:

The **sine wave** is the most fundamental form of a periodic analog signal. It is as simple oscillating curve, its change over the course of a cycle is smooth and consistent, a continuous rolling flow.

The sine wave can be fully described by three characteristics: amplitude, period or frequency and phase.

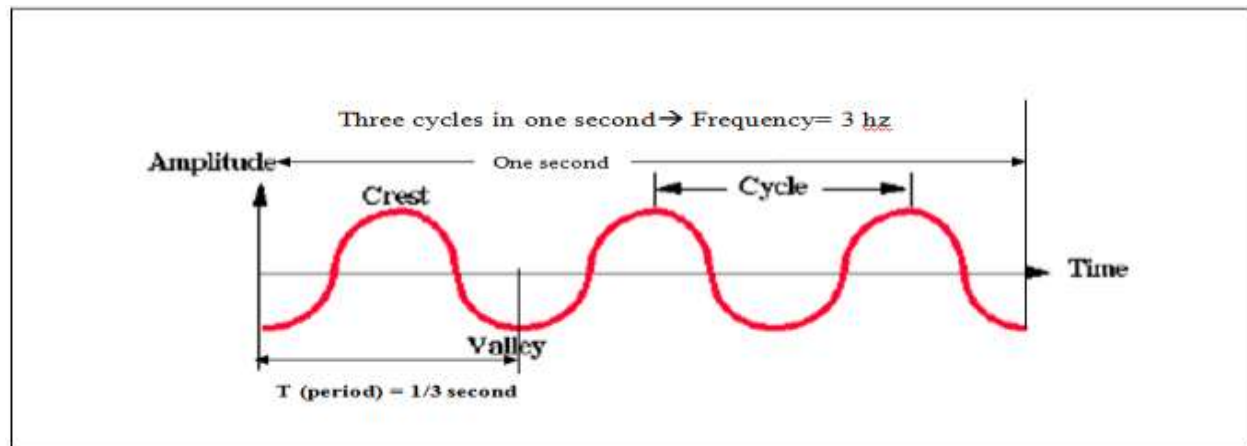
i) Amplitude:

On a graph, the amplitude of a signal is the value of the signal at any point on the wave. It is equal to the vertical distance from a given point on the wave form to the horizontal axis. The maximum amplitude of a sine wave is equal to the highest value it reaches on the vertical axis.



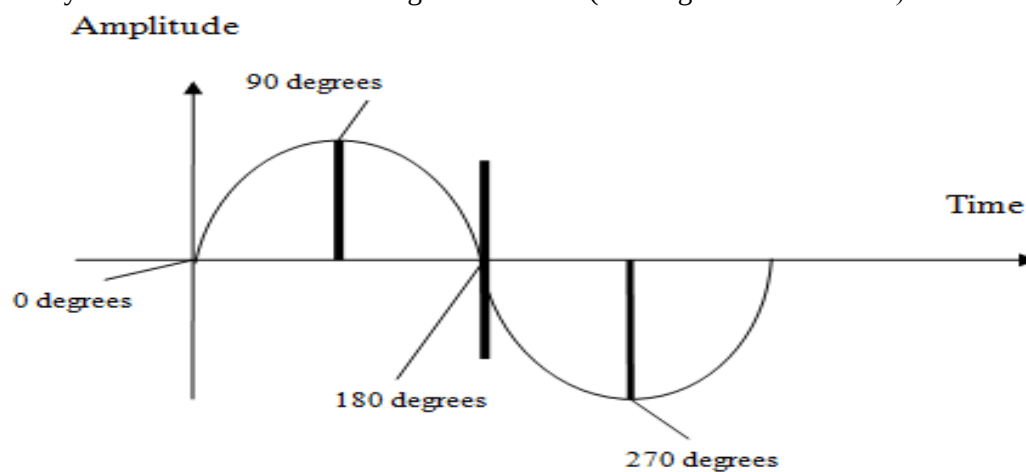
ii) Period or Frequency:

Period refers to the amount of time, in seconds, a signal needs to complete one cycle. Frequency refers to the number of periods in one second. So the **frequency of a signal is its number of cycles per second**.



iii) Phase:

The term phase describes the position of the wave form relative to time zero. Phase indicates the status of the first cycle. Phase is measured in degree or radians (360 degrees is 2π radians).



Encoding and modulating

We must transform the data into signals to send them from one place to another. How the information is transformed depends on its original format and on the format used by the communication hardware.

Data Encoding

Encoding is the process of converting data from one form to another.

Digital data to digital signal Digital data to digital signal encoding may be used as the equipment for encoding digital data into a digital signal. Eg: computer communication using digital line (eg. ISDN)

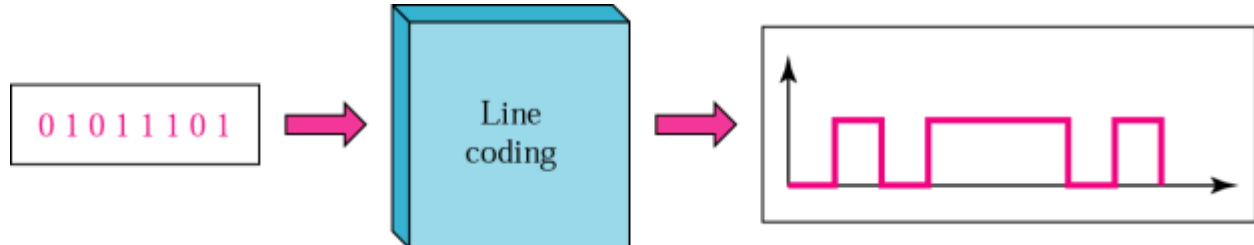
Digital data to analog signal may be used as some transmission media, such as optical fibre and unguided media, will only propagate analog signals Eg: computer communication using voice (telephone) line.

Analog data to analog signal Analog data and analog signal combination will be used as analog data in electrical form can be transmitted as baseband signals easily and cheaply. Eg: home telephone systems.

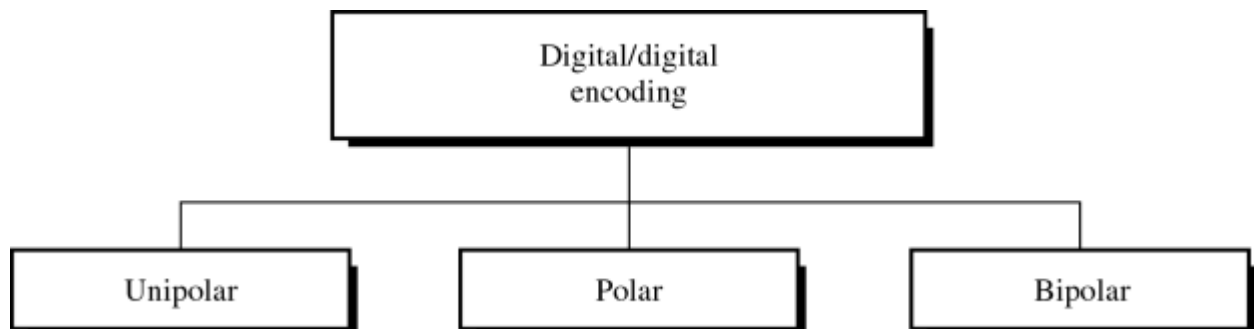
Analog data to digital signal Analog data to digital signal conversion permits the use of modern digital transmission and switching equipment. Eg: digital mobile communication systems.

1. Digital to digital encoding

Digital to digital encoding or conversion is the representation of digital information by a digital signal. In this type of encoding the binary 1 and 0 are translated into a sequence of voltage pulses that can be transmitted over wire.

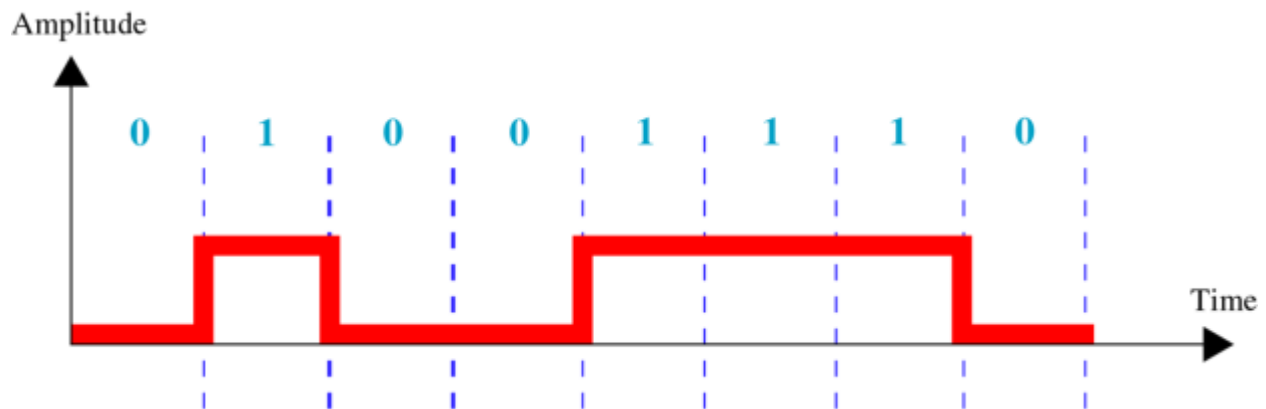


Types of Digital to Digital Encoding



1.1 Unipolar :-

It uses only one level of value and polarity . Digital transmission system works by sending voltage pulses along a medium link usually cable or wire. The polarity of a pulse refers to whether it is positive or negative.



In this example the 1's are encoded as a positive value and the 0's are encoded as the negative value.

Advantages :-

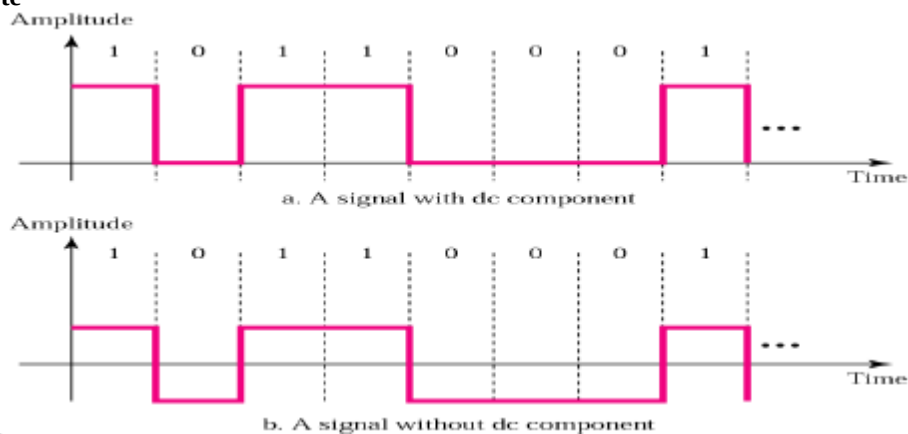
- Very simple.

Disadvantages :-

- DC component.
- Synchronization.

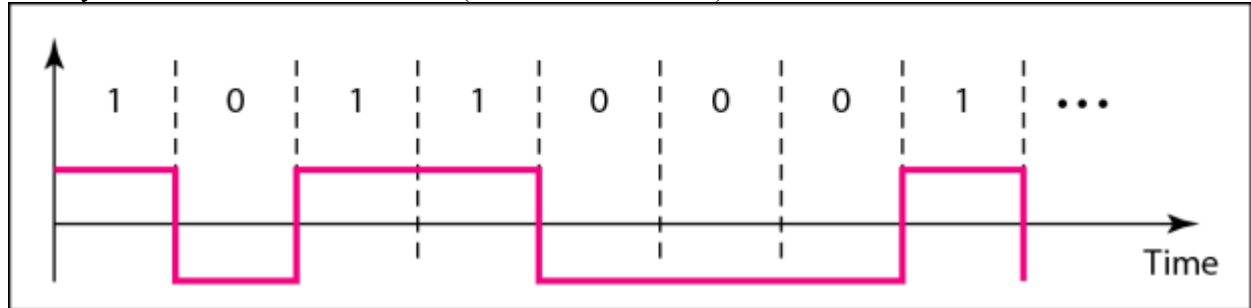
DC component :- When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies, called DC components, that present problems for a system that cannot pass low frequencies." Dc is extra energy residing on the line and is useless.

Note

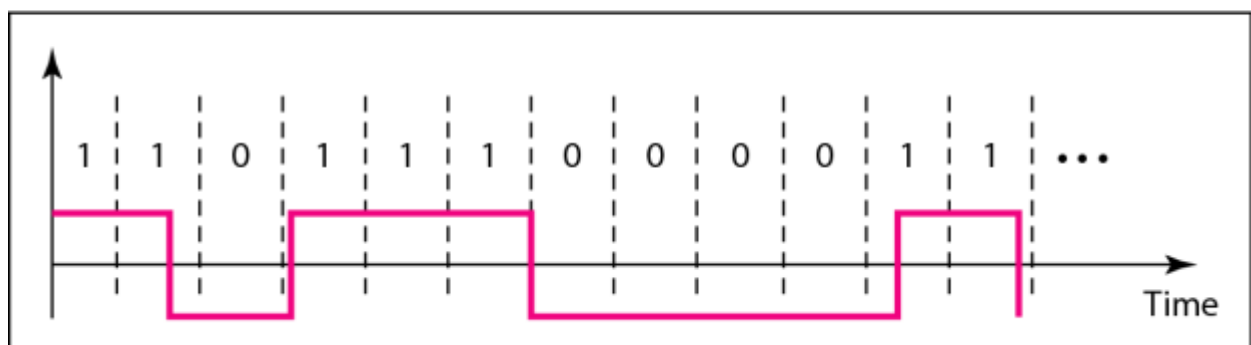


Synchronization :- when a signal is unvarying, the receiver cannot determine the beginning and ending of each bit. Therefore, a synchronization problem in unipolar can occur when ever the data stream includes a long uninterrupted series of 1's or 0's. Thereceiver receives a continuous voltage and receiver determines how many bits are sent by relaying on its clock (bitrelaying bit--duration), which may not be synchronized with the sender clock.

To correctly interpret the signals received , the receiver's bit intervals must correspond exactly to the sender's bit intervals (the same clock rate).



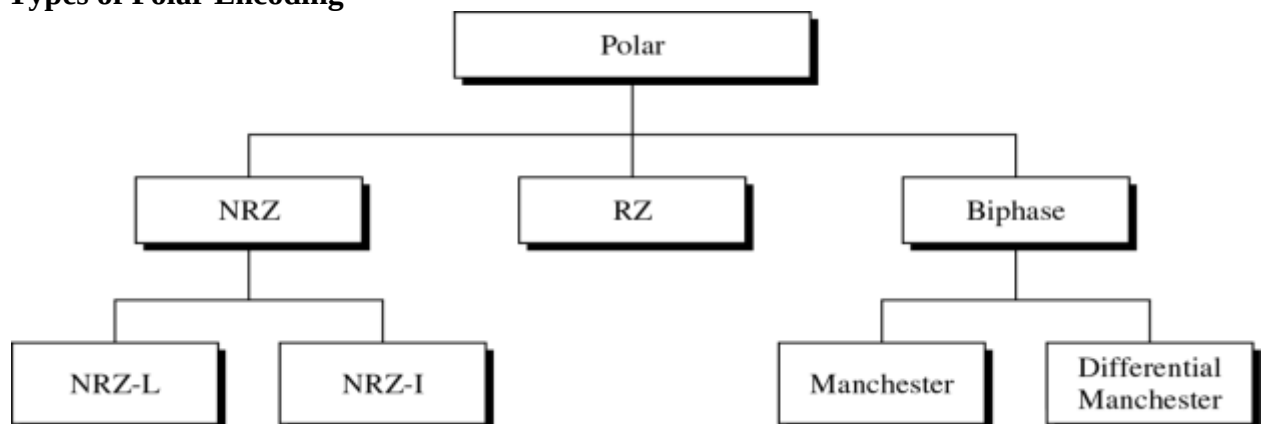
a. Sent



b. Received

1.2 Polar :- It uses two levels of amplitude. One is positive and another is negative, so the DC component problem of unipolar encoding is eliminated.

Types of Polar Encoding



In this encoding technique, the level of the signal is always either positive and negative. It has two types: NRZ-Level, NRZ-Invert.

Non return to zero(NRZ)

NRZ –L :- In NRZ-L the level of the signal is dependent upon the state of the bit. A positive voltage usually means the bit is a zero, and a negative voltage means the bit is one or (vice-versa).

Advantages-Eliminate DC component.

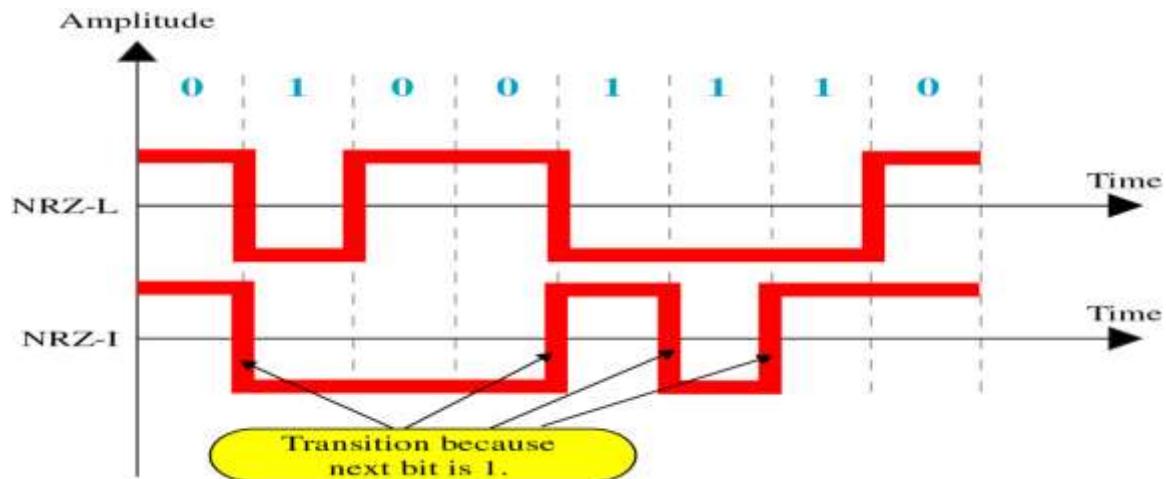
Disadvantages-Lack of synchronization.

Non return to zero(NRZ)

NRZ-I :- In this the signal is inverted if a one is encountered and zero bit is represented by no change.

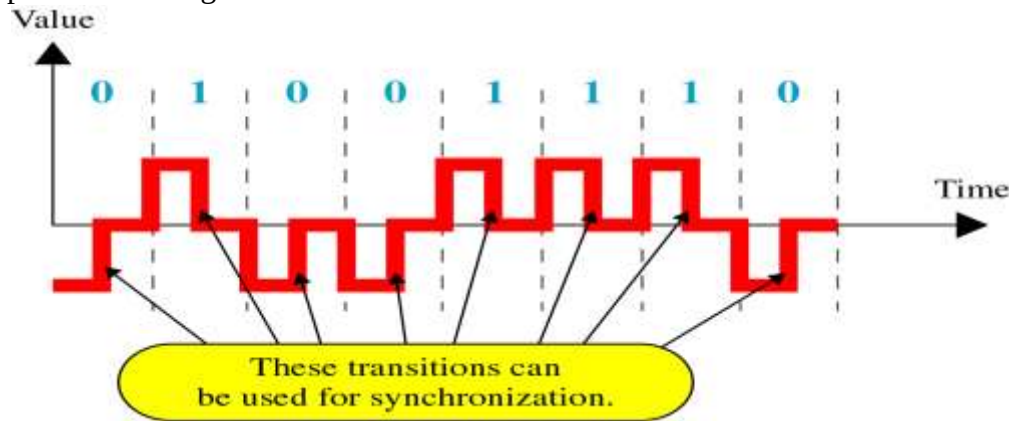
Advantages- Synchronization provided by the signal change.

Disadvantages- String of zeros can still cause problems.



Return to zero (RZ) :- RZ uses three values: positive, negative and zero. In this, the signal changes not between bits but during each bit. A positive voltage means one and negative voltage means zero and halfway through each bit interval, the signal returns to zero.

one bit is represented by positive-to-zero and a **zero bit** by negative-to-zero, rather than by positive and negative alone.



Advantages :-

- Better synchronization.

Disadvantage :-

- It requires two signal changes to encode one bit and therefore occupies more bandwidth.

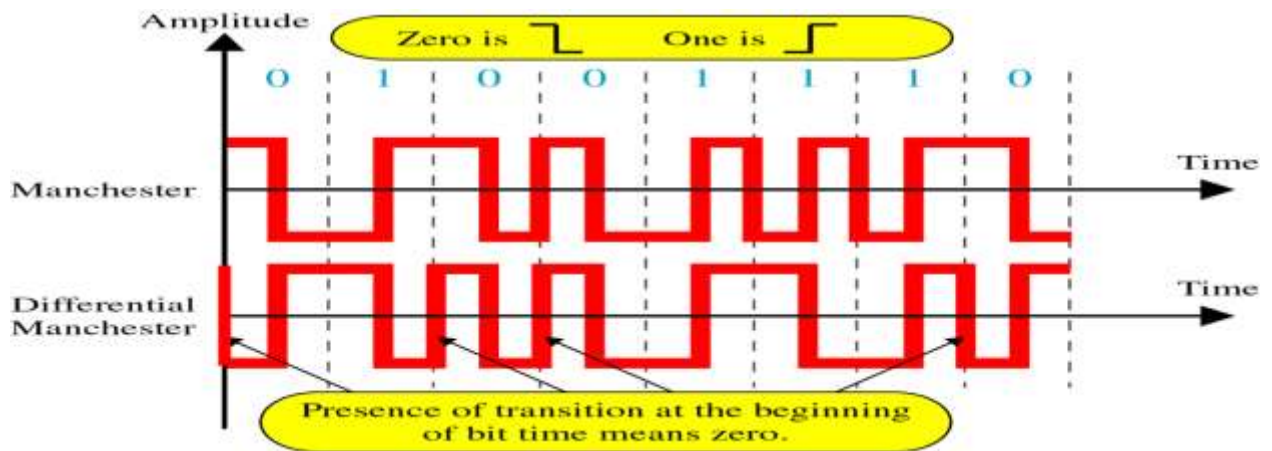
Biphase

In this method, the signal changes at the middle of the bit interval but does not return to zero. It can be implemented in two ways:-

- **Manchester**
- **Differential Manchester.**

Manchester encoding:- It uses the inversion at the middle of each bit interval for both synchronization and bit representation. A **negative to positive translation** represent **binary one** and a **positive to negative translation** represent **binary zero**. By using the single translation for a dual purpose, manchester encoding achieves the same level of synchronization as RZ but with only two levels of amplitude.

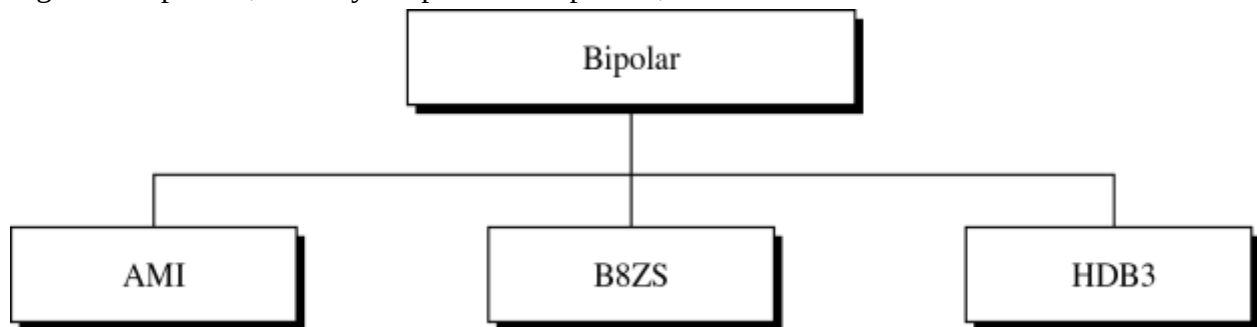
Differential Manchester:- In this, the transition at the middle of the bit is used only for synchronization. The bit representation is shown by the inversion or non inversion at the beginning of the bit. A transition means binary 0 and no transition means binary 1. It requires two signal changes to represent binary 0 but only one to represent binary 1.



1.3 **Bipolar encoding** :-

It uses three levels of voltage or amplitude: positive, negative and zero. The zero level is used to represent binary zero. The 1's are represented by alternative positive and negative voltages.

If the first one bit is represented by the positive amplitude, second will be represented by negative amplitude, third by the positive amplitude, and so on.

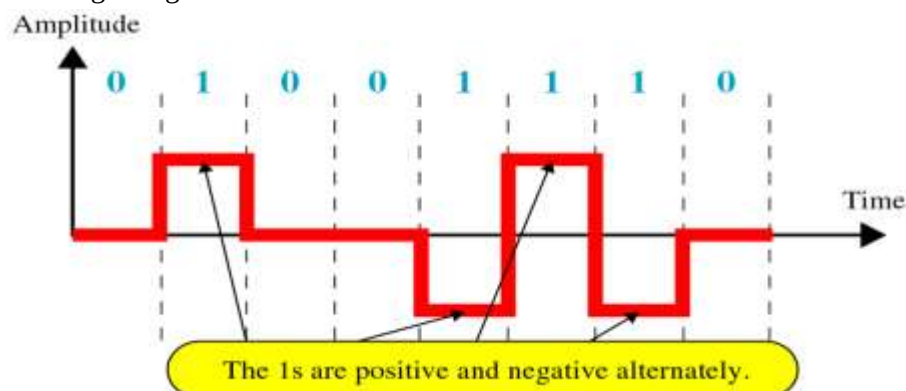


Alternate mark inversion :-

Mark means one, so AMI means alternate one inversion. zero voltage represent binary zero and 1's are represented by alternating positive and negative voltages.

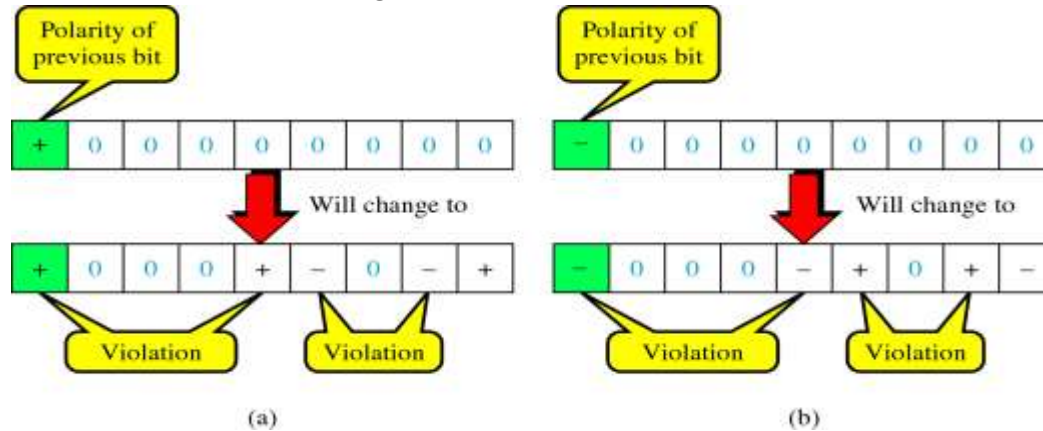
AMI accomplishes two things:

1. DC component is zero.
2. Long sequence of 1s stays synchronization but there is no mechanism to ensure the synchronization of a long string of 0s.



Bipolar 8 zero substitution:-

In this if 8 zeros come one after another, we change the pattern in one of two ways based on the polarity of the previous. The difference between B8ZS and AMI occurs whenever eight or more consecutive 0s are encountered in the data stream. Solution provided by B8ZS is to force artificial signal changes, called violations, within the zero string.



If the previous 1 bit was positive, the eight 0s will be encoded as zero, zero, zero, positive, negative, zero, negative, positive.

When receiver finds two consecutive positive changes surrounding three 0s, it recognizes the pattern as deliberately introduced violation and not an error. It then looks for the 2nd pair of the expected violations. When it finds them, the receiver translates all eight bits to 0s and reverts back to normal bipolar AMI mode. If the polarity of the previous 1 bit is negative, then pattern of violation will be same but with inverted polarities.

High density bipolar-3 :-

In this if four zeros come one after another, we change the pattern based on the polarity of the previous 1 and the number of 1s since the last substitution.

When the no. of 1s since the last substitution is odd, HDB3 Puts a violation in the places of 4th consecutive zeros.

if the polarity of the previous bit was positive, then the violation is positive.

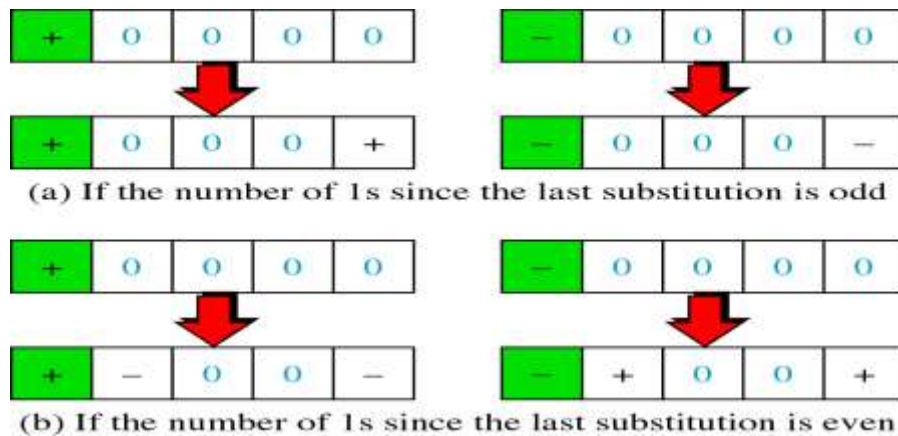
if the polarity of the previous bit was negative, then the violation is negative.

When the no. of 1s since the last substitution is even, HDB3 Puts violations in the places of both the first and the fourth consecutive zeros.

if the polarity of the previous bit was positive, both violations are negative.

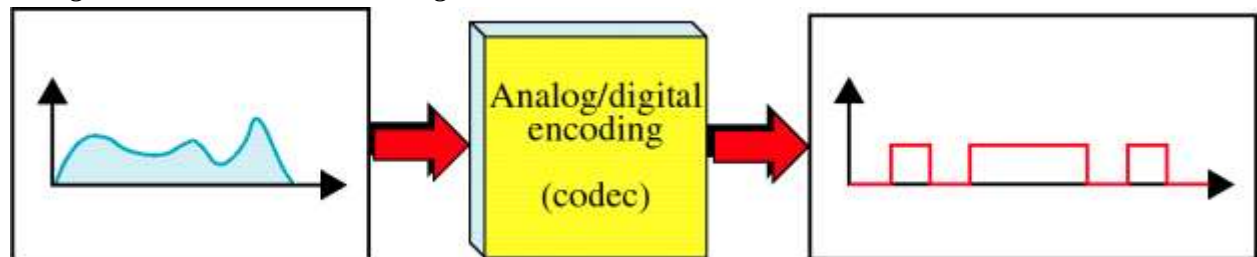
if the polarity of the previous bit was negative, both violations are positive.

This was introduced by Europe and Japan and earlier was in US.



2. Analog to Digital Encoding

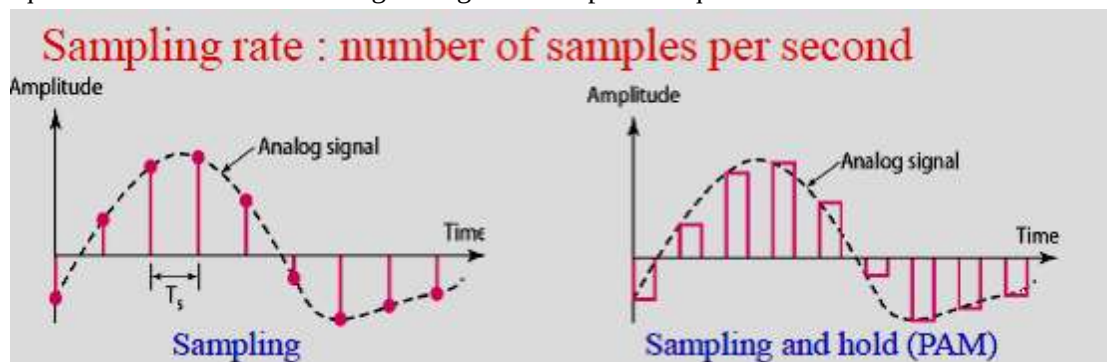
The idea of digitizing analog signal started with telephone system, to provide long distance services. They digitized the analog signals at the sender. The digital signal thus consists of blocks of n bits. On reception the signal is converted back to analog at the receiver end.



The problem here is how to translate information from an infinite no of values to a discrete no of values without sacrificing sense or quality.

2.1 PAM (Pulse Amplitude Modulation)

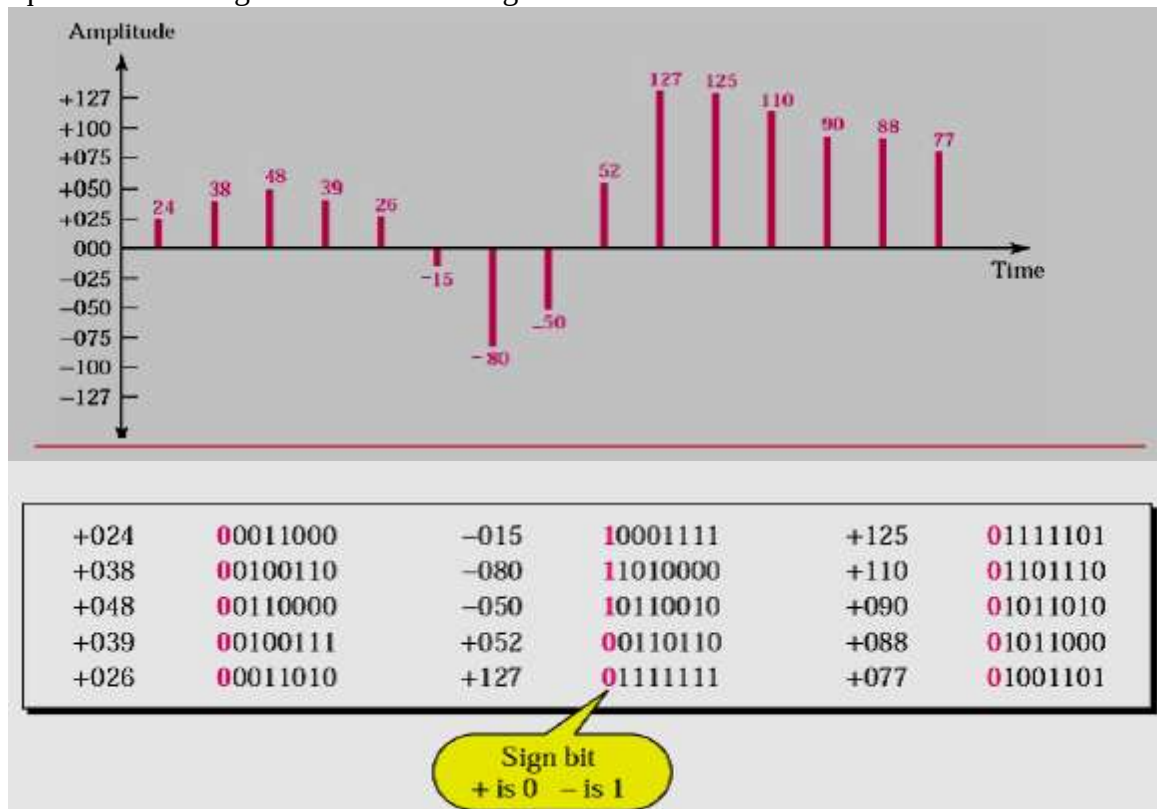
This is the first step as this technique takes an analog signal, samples it and generates a series of pulses based on the result of the sampling. The term sampling means measuring the amplitude of the signal at equal intervals. In PAM the original signal is sampled at equal intervals.



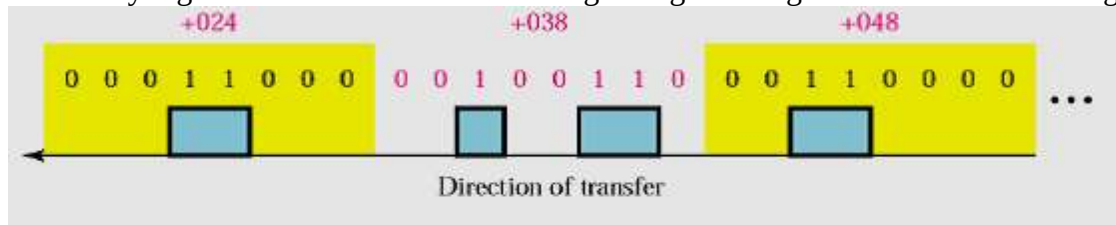
The reason PAM is not useful to data communication is that it translates original wave form to a series of pulses, which are of any amplitude.

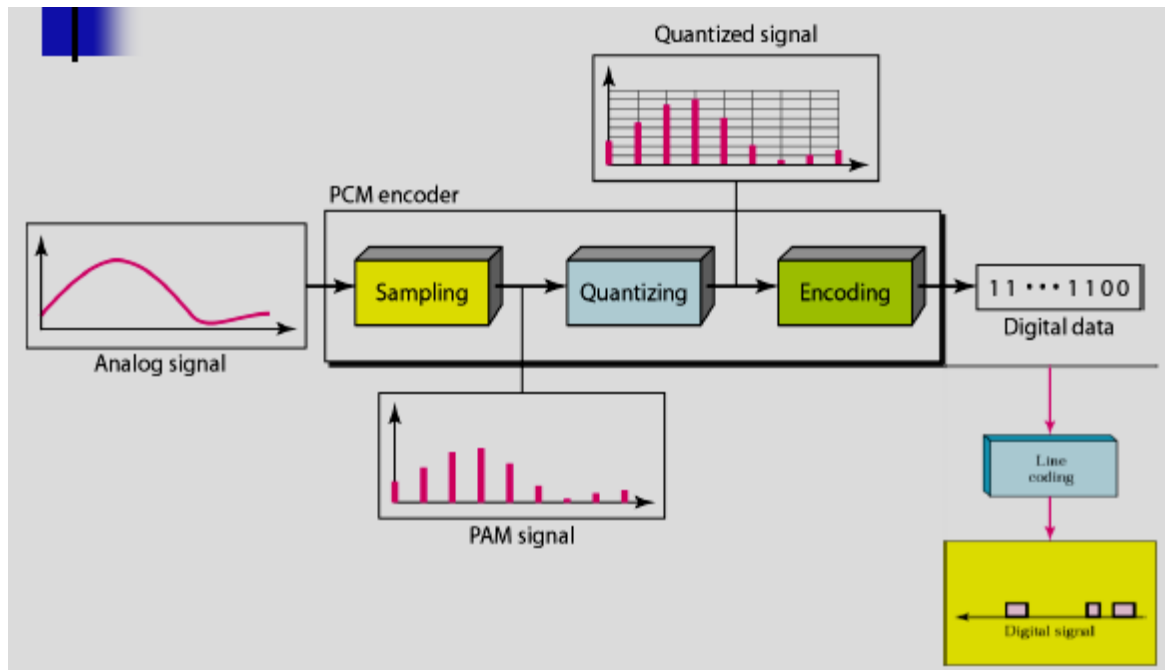
2.1 PCM (Pulse Code Modulation)

PCM modifies the pulses created by PAM to create a completely digital signal. So the first step is PCM quantizes the PAM pulses. Quantization is a method of assigning integral(magnitude) values in a specific ranges to sampled instances. Each value is translated into its seven-bit binary equivalent. The eight bit indicate the sign.



The binary digits are then transformed to a digital signal using one of the line encoding.

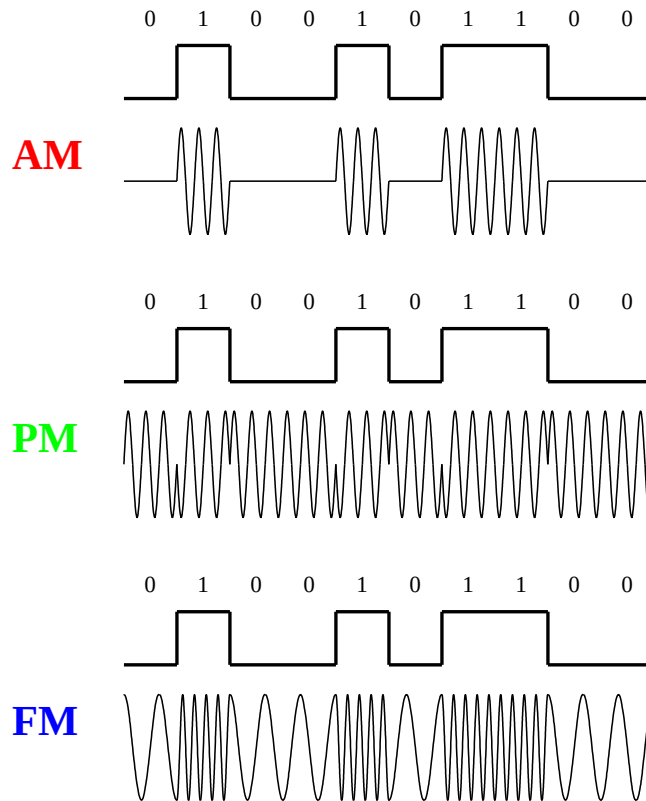




3 Digital data to analog signals

A modem (modulator-demodulator) converts digital data to analog signal. There are 3 ways to modulate a digital signal on an analog carrier signal.

1. **Amplitude shift keying (ASK):** is a form of modulation which represents digital data as variations in the amplitude of a carrier wave. Two different amplitudes of carrier frequency represent '0' , '1'.
2. **Frequency shift keying (FSK):** In Frequency Shift Keying, the change in frequency define different digits. Two different frequencies near carrier frequency represent '0' , '1'.
3. **Phase shift keying (PSK):** The phase of the carrier is discretely varied in relation either to a reference phase or to the phase of the immediately preceding signal element, in accordance with data being transmitted. Phase of carrier signal is shifted to represent '0' , '1'.



Analog to analog signals

It is the representation of analog information by an analog signal. Modulation has been defined as the process of combining an input signal $m(t)$ and a carrier at frequency f_c to produce a signal whose bandwidth is usually centred on f_c . Analog to analog modulation can be done in three ways:

4.1 Amplitude modulation (AM)

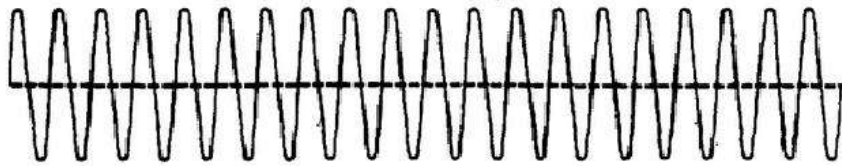
In AM transmission, the carrier signal is modulated so that its amplitude varies with the changing amplitudes (voltage level) of the modulating signal. The frequency and phase of the carrier remain the same; only the amplitude changes to follow variations in the information.

4.2 Frequency modulation (FM)

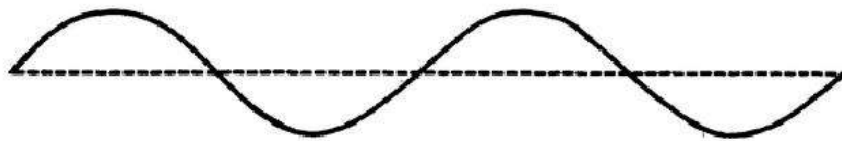
In FM transmission, the carrier signal is modulated so that its frequency varies with the changing amplitudes (voltage level) of the modulating signal. The peak amplitude and phase of the carrier remain the same; only the frequency changes to follow variations in the information.

4.3 Phase modulation (PM)

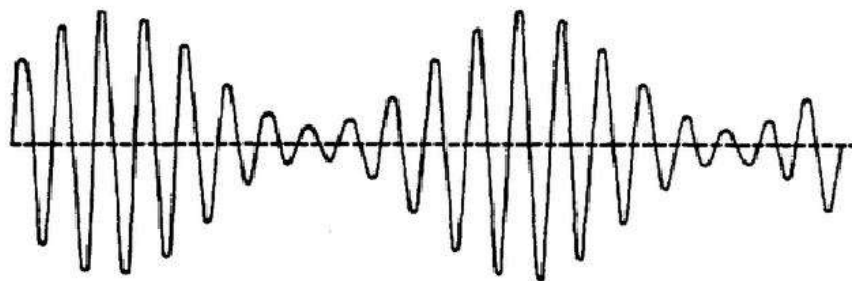
Due to simpler hardware requirements, PM is used in some systems as an alternative to FM. In PM transmission, the carrier signal is modulated so that its phase varies with the changing amplitudes (voltage level) of the modulating signal. The peak amplitude and frequency of the carrier remain the same; only the phase changes to follow variations in the information. The analysis and the final result are similar to FM.



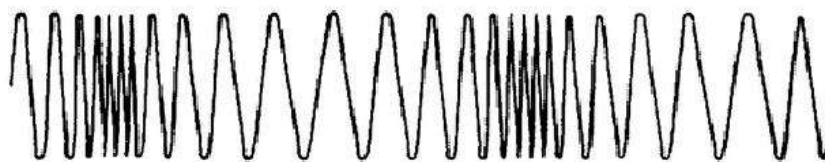
A CARRIER



B SIGNAL



C AMPLITUDE MODULATED WAVE



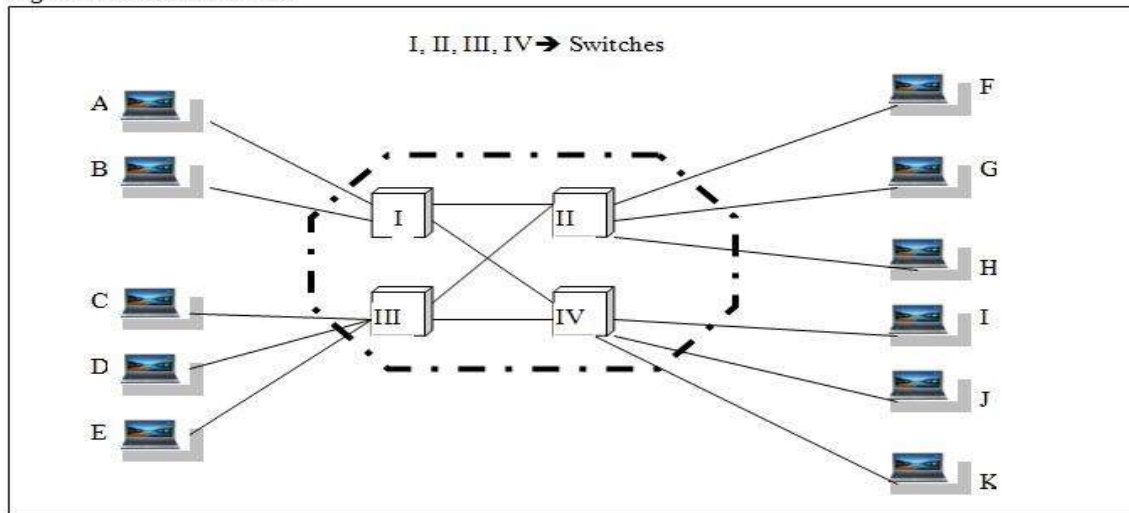
D FREQUENCY MODULATED WAVE

Switching:

Whenever we have multiple devices to connect with each other, the better solution is switching. A switched network consists of a series of interlinked nodes called **switches**.

Switches are hardware/software devices that are capable of creating the temporary connections between two or more devices link to the switch.

Figure: *Switched Network*

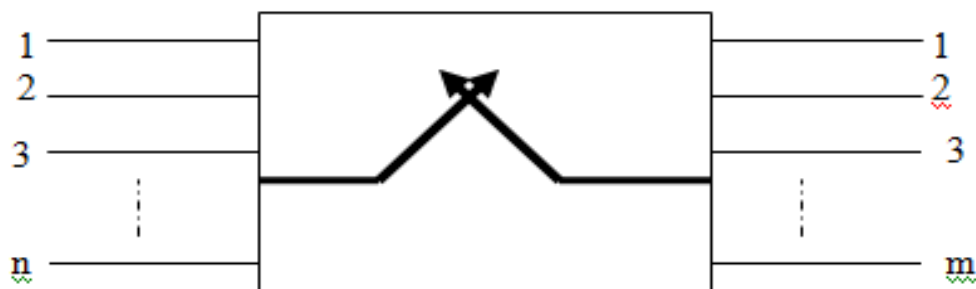


There are three methods for switching.

- I) Circuit switching
- II) Packet Switching
- iii) Message Switching

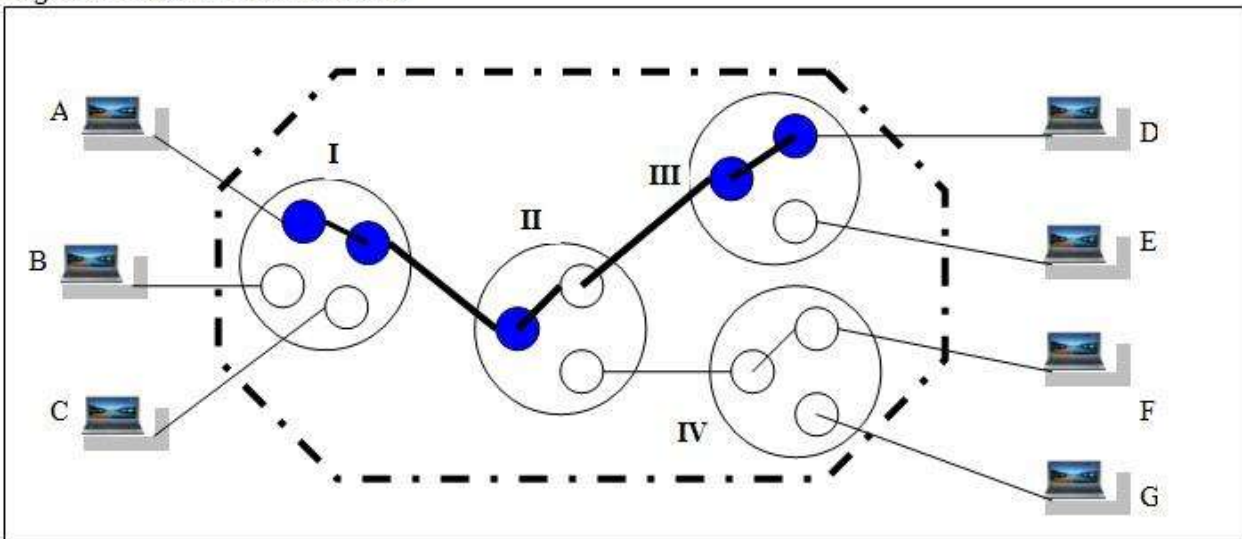
I) Circuit Switching:

Circuit switching creates a direct physical connection between two devices such as phones. A circuit switch is a device with n inputs and m outputs that creates a temporary connection between an input link and an output link. The number of inputs may have different with the number of outputs.



A Circuit Switch

Figure: *Circuit-Switched Network*



Circuit Switching use either of two technologies:

- 1) Space-Division Switches
- 2) Time-Division Switches

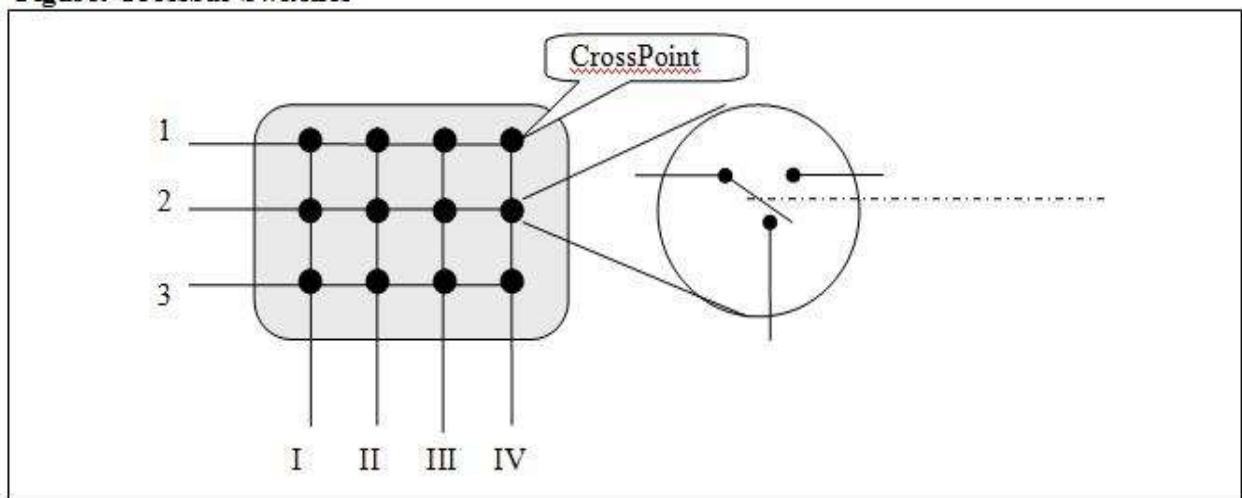
1) Space-Division Switches:

In space-division switching, the path for each communication in the circuit are separated from each other. There are many technologies used to design the **Space Division Switches**.

a) Crossbar Switches:

A **crossbar switch** connects n inputs to m outputs in a grid using transistor at each cross point. So for connecting n inputs and m outputs there are $n \times m$ cross points required.

Figure: Crossbar Switches

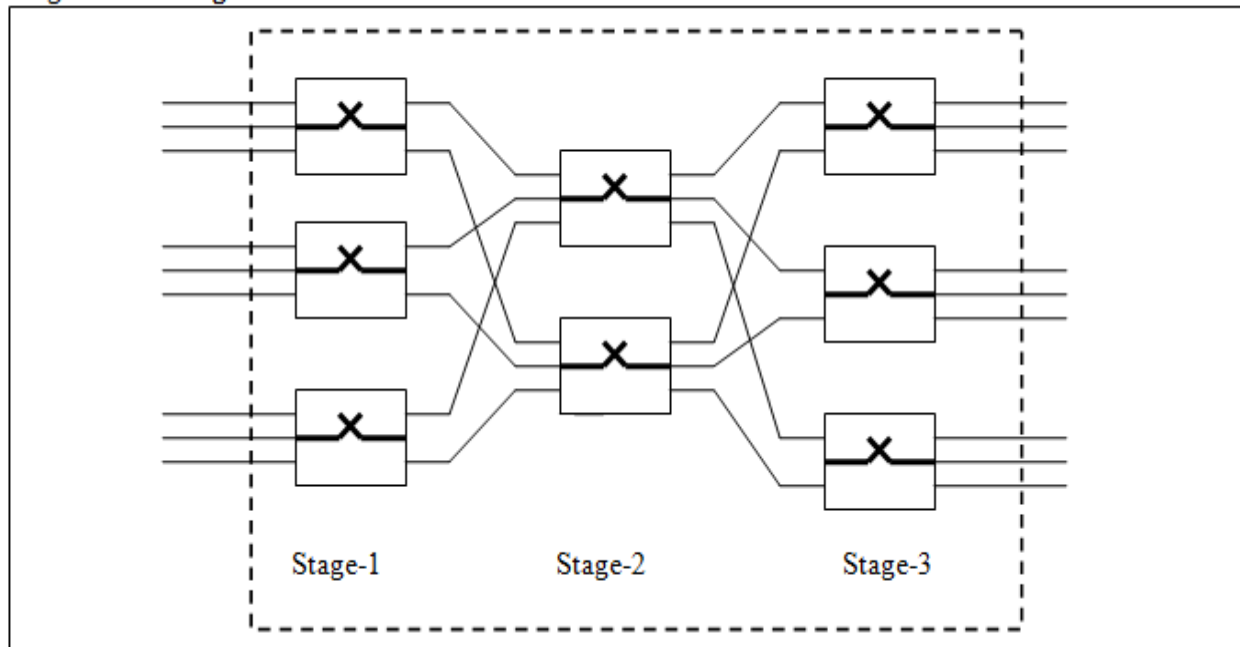


This factor makes the cross impractical because readings show that in practice fewer than 25% of the cross points are in use at a given time. The rest are ideal.

b) Multistage Switches:

The solution to reduce the number of the crossbar switch is to use multistage switches, which combine crossbar switches in several stages.

Figure: *Multistage Switch*



The design of multistage switch depends on the number of stages and the number of switches requires in each stage. Normally middle stages have fewer switches than the first and last stage. Let us compare the number of cross points in a 9-by-9 single stage crossbar switch with the 15-by-15 multistage switch.

In a single stage switch, we need 81 cross points (9×9).

In the multistage switch, we need

- ❖ Three, first stage switches, each with 6 cross points (3×2), total of 18 cross-points at the first stage.
- ❖ Two, second stage switches, each with 9 cross points (3×3), total of 18 cross-points at the first stage.
- ❖ Three third stage switches, each with 6 cross points (3×2), total of 18 cross-points at the first stage.

So in multistage switching we need only 54 total crosspoints which is 34% less than single stage switching.

2) Time-Division Switches:

Time-division switching uses time-division multiplexing to achieve switching. There are two popular methods used in time-division multiplexing:

- a) Time-slot Interchange
- b) TDM Bus

a) Time-Slot Interchange (TSI):

In **TSI**, we insert a device called **Time-slot interchange (TSI)**. A TSI changes the ordering of the slots based on the desired connections. In the figure, it changes the order of data from A, B, C, D to C, D, A, B. Now at the receiver end demultiplexer separates the slots, it passes them to the proper output.

Figure: *Time-Division Multiplexing without TSI (No Switching)*

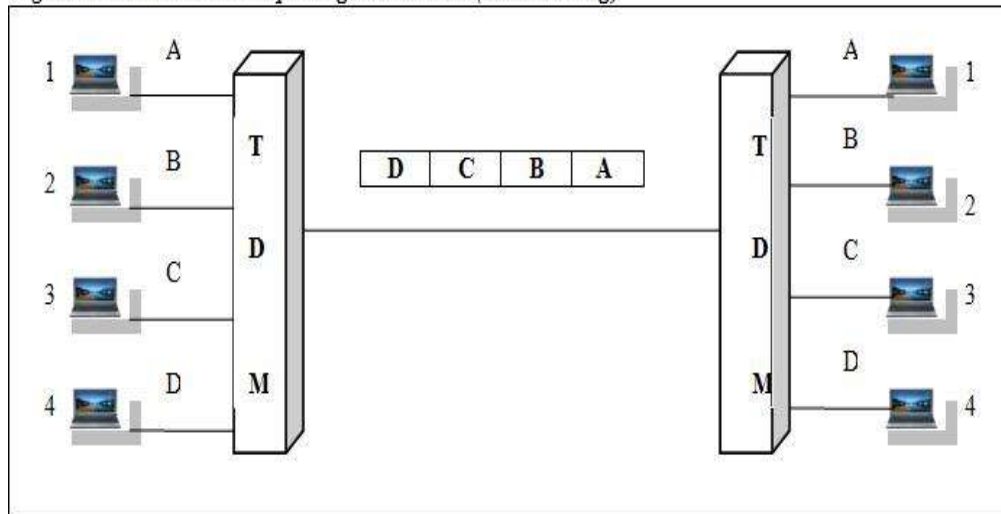
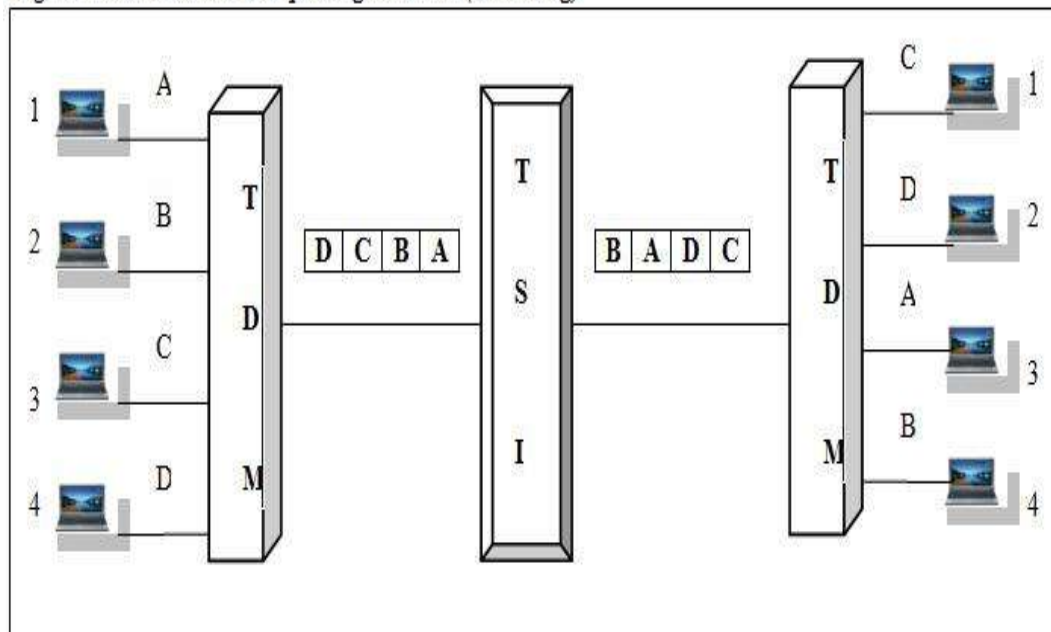


Figure: *Time-Division Multiplexing with TSI (Switching)*



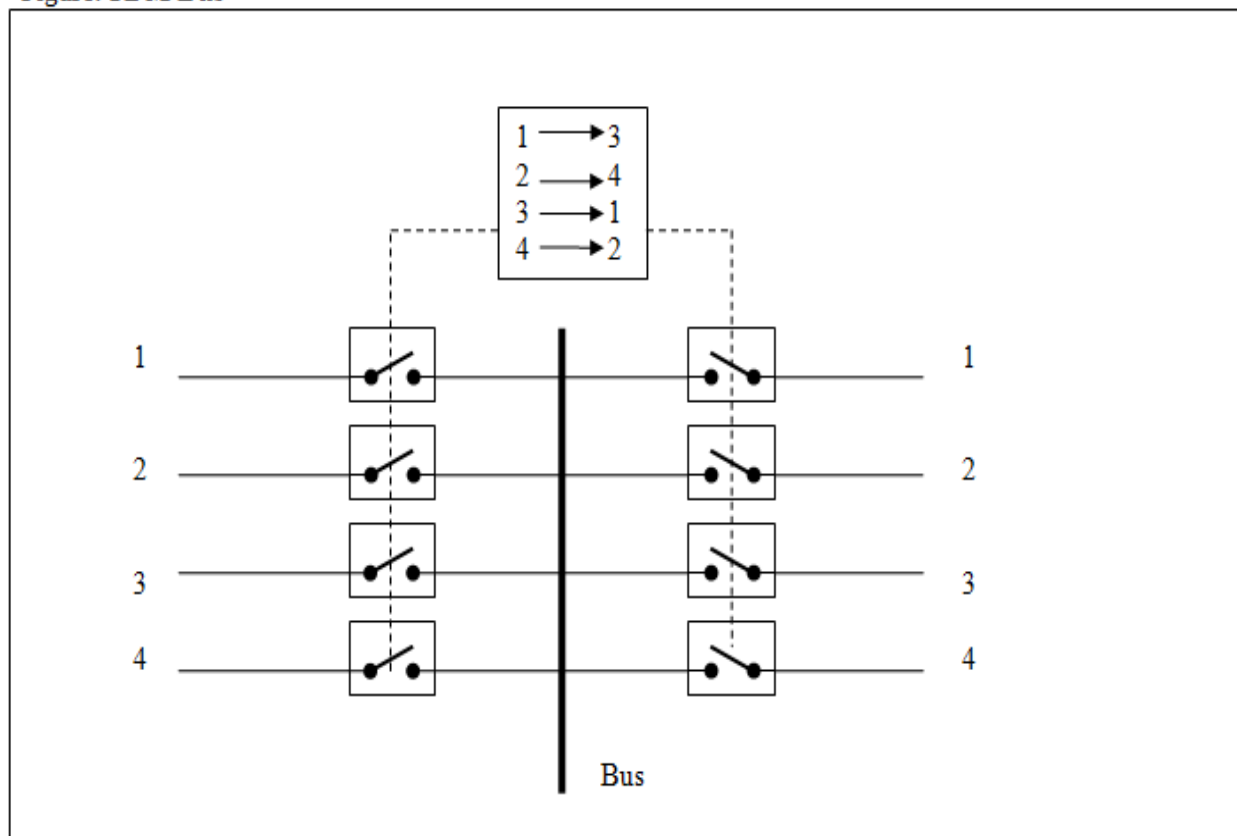
TSI consists of RAM (random access memory) with several memory locations. The size of each location is the same as the size of a single time slot. The RAM fills up with incoming data from

time slots in the order received. Slots are then sent out in an order based on the decisions of control unit.

b) TDM Bus:

In **TDM Bus**, the input and output lines are connected to a high speed bus through input and output gates. During the data transfer one input gate is closed for a time slots, during the same time slot, only one output gate is also closed. This pair of gates allows transferring of data from one specific input line to one specific output line using bus. The control unit opens and closes the gate accordingly to switching need. For example: in the figure, at the first time slots the input gate 1 and output gate 3 will be closed; during the second time slot, input gate 2 and output gate 4 will be closed; and so on.

Figure: *TDM Bus*



II) Packet Switching:

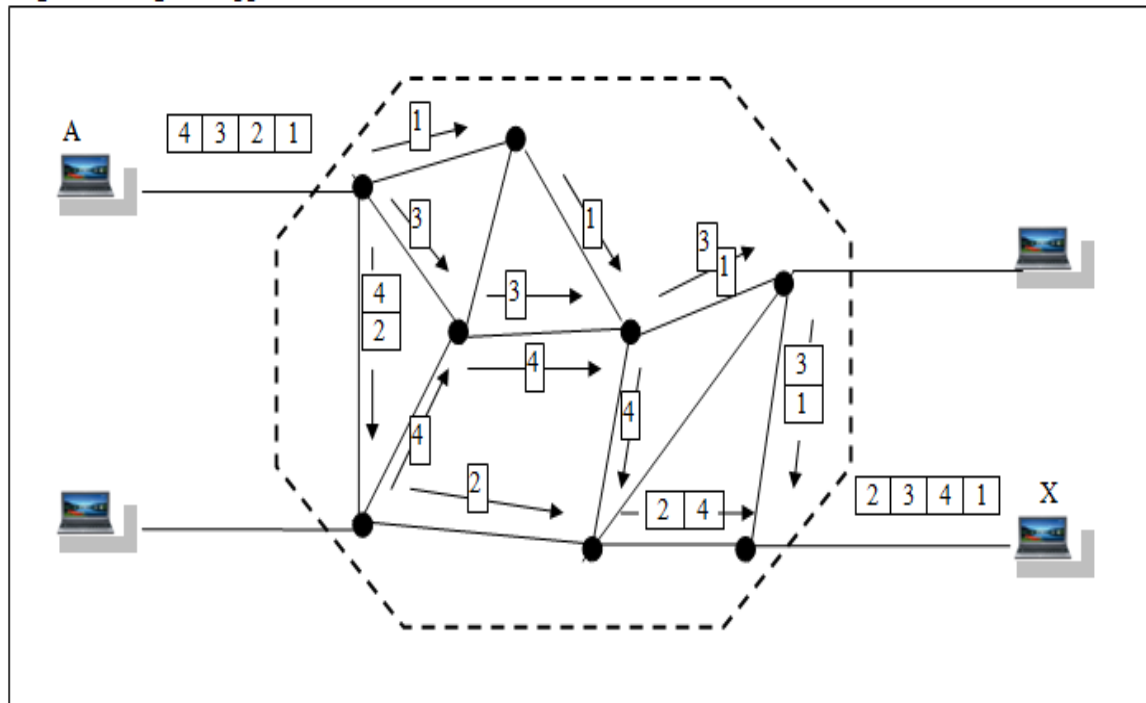
In a **packet switched network**, data are transmitted in discrete units of variable length blocks called **packets**. Longer transmissions are broken up in to multiple packets. Each packet contains not only data but also a header with control information (such as priority codes, source and destination address). The packets are sent over the network node to node. At each node, the packet is stored then routed according to the information in its header. There are two popular approaches for packet switching

- 1) Datagram Approach
- 2) Virtual Circuit Approach

1) Datagram Approach:

In **datagram approach**, each packet is treated independently from all others. Packets in this technology are referred to as **datagram**. In this approach the datagram reach to their destination out of order. It is the responsibility of the transport layer to reorder the datagram before passing them to destination port.

Figure: Datagram Approach



Virtual

2) Virtual Circuit Approach:

In **virtual circuit approach**, the relationship between all packets belonging to a message or session is preserved. A single route is chosen between sender and receiver at the beginning of the session. When data are sent, all packets of the transmission travel one after another along that route only.

III) Message Switching:

In **message switching**, a node receive a message, store it until the appropriate route is free and then send forward.

Note that in message switching, the message are stored and relayed from secondary storage (disk), while in packet switching the packet are stored and forwarded from primary storage (RAM).

Figure: Message Switching

